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About this copy

This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. N] shows where page N of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM14.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Proof of (3.1), page 28: the proof is added.
2. **Statement mistake.** Section 3, page 28: in the definition of a λ-zone, "hence" is removed, so that "$\overline{\Lambda}$ is a closed disc" becomes a condition.
3. **Missing proof.** Proof of (5.1), page 31: the proof is added.
4. **Missing proof.** Proof of (5.2), page 31: the proof that $\mathscr{R}$ is connected, and that the disc includes Z in its interior, is added.
5. **Missing proof.** Proof of (5.5), page 32: the proof is added.
6. **Misprint.** Proof of (6.5), page 35: a missing α is inserted, so that the text reads $bd(\alpha(A,\Omega))$.
7. **Proof gap.** Proof of (6.5), page 35: the discs $\alpha(A,\Omega)$ are first moved away from $bd(\Delta)$.

8. **Proof gap.** Proof of (7.3), page 36: the step $1 \geq d(z, v)$ is removed.
9. **Missing proof.** Proof of (8.3), page 41: the proof is added.
10. **Missing proof.** Claim (1) in the proof of (8.4), page 41: the replacement of the path P is added.
11. **Proof gap.** Claim (2) in the proof of (8.4), page 42: $B \setminus w$ becomes $A \setminus w$, and a note explains a gap.
12. **Proof gap.** Proof of (8.4), page 42: a note explains how to apply (8.2) when the paths end on C_2.
13. **Missing proof.** Proof of (9.3), page 45: the step that contradicts the maximality of Λ is added.
14. **Misprint.** Section 9, page 45: in clause (v) of the definition of a battlefield, $A(K)$ becomes $A(H)$.

[p. 23] # Graph Minors
XIV. Extending an Embedding

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Let G be a graph, and let H be a subgraph of G drawn in a surface Σ. When can this drawing be extended to an embedding of the whole of G in Σ, up to 3-separations? We show that if such an extension is impossible, and if H is a subdivision of a simple 3-connected graph and is highly "representative", then one of two obstructions is present. This is a lemma for use in a future paper.

1. Introduction

The next goal of this series of papers is to show that, for any fixed graph H, all graphs with no minor isomorphic to H are tree-structures of pieces which in some sense are "almost" of low genus. We are therefore concerned with theorems saying that graphs not possessing certain obstructions can be embedded in simple surfaces. One natural question is, given a graph G and a surface Σ, when can G be embedded in Σ? But answering this in a way which is useful for proving theorems (rather than an algorithmic answer) is probably close to finding explicitly all the excluded minors for embedding in Σ, and is therefore much too difficult. Fortunately, we do not really need to answer this question. In our application, we already have a subgraph H of G drawn in Σ which is highly "representative" (that is, every non-null-homotopic closed curve in Σ meets the drawing of H a large number of times), and it suffices for us to find a condition under which this [p. 24] drawing of H can be extended to a drawing of G. This is a much easier question. Every vertex of H is surrounded by a "large planar neighbourhood" in Σ, and the remaining parts of G attached in this neighbourhood have to be drawn within it, if the extension is possible; and

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so the problem essentially becomes a problem about extending drawings in discs. In fact, we can make the problem even easier, because we are only concerned with the case when H is a subdivision of a simple 3-connected graph. The advantage of this hypothesis is that we get better, more useful, obstructions, because we cannot have "made a mistake" in the embedding of H. If this embedding of H does not extend to one for all of G, we cannot locally redraw H so that the embedding does extend, because there are no other ways to draw H (this is a consequence of a theorem of Whitney, as we shall see); and this makes the obstructions healthier, less trivial.

On the other hand, one obstruction might be an edge e of G not in H, with ends $u, v \in V(H)$, not in the same region, but lying in distinct regions of H which are incident with a common edge f. If we draw e in the natural way crossing f, we get a graph drawn in Σ with one crossing. In our application we can only make use of such a graph if the four ends of e, f are not cut off from the main part of H by a 3-separation. Crossings which are cut off by 3-separations are of no use to us, and we must modify the definition of "extending the embedding of H" so that these no longer are obstructions. For this reason, we shall only be concerned with the question, when can the embedding of H be extended to one of G "up to 3-separations"? (We shall make this precise later.) Then all the obstructions (there are only two) turn out to be objects that we can make use of in our application.

In order to give the reader a better idea of what we are aiming for, we first state a simplified special case of the result, as follows.

(1.1) *Let G be a 4-connected graph, and let H be a subgraph of G which is a subdivision of a simple 3-connected graph. Let Σ be a connected surface without boundary, not a sphere, and let H be drawn in Σ. Suppose that every closed curve in Σ that meets H at most 208 times is null-homotopic. For such a curve ϕ, let* $\mathrm{ins}(\phi)$ *denote the union of the points of Σ appearing in ϕ together with all discs in Σ with boundary in this set. Then either*

(i) *the drawing of H can be extended to one of G, or*

(ii) *there is a path of G with distinct ends $u, v \in V(H)$ and with no other vertex or edge in H, such that there is no closed curve ϕ meeting H at most 3 times with $u, v \in \mathrm{ins}(\phi)$, or*

(iii) *there is a circuit C of H bounding an open disc $\Lambda \subseteq \Sigma$, and a point x of Λ, and four vertices a, b, c, d of C in order, such that*

[p. 25] (i) *$H \cap (\Sigma - \Lambda)$ is a subdivision of a simple 3-connected graph*

(ii) *for every $y \in \Lambda$ there is a closed curve ϕ meeting H at most 39 times, with $\{x, y\} \subseteq \mathrm{ins}(\phi)$*

(iii) $\mathrm{bd}(\Delta) - \Lambda$ *meets H at least 4 times, for every closed disc $\Delta \subseteq \Sigma$*

with $a, b, c, d \in \Delta$, *and*

(iv) *there are two disjoint paths of* G *with ends* a, c *and* b, d *each meeting* H *only within* $\Lambda \cup \{a, b, c, d\}$.

This simplified form is not very far from what we really want. The main changes we need are:

(i) we need to permit Σ to be a sphere; and then the definition of $\mathrm{ins}(\phi)$ becomes more complicated, using tangles

(ii) G might not be 4-connected; then we ask (in (1.1)(i)) only that the drawing be extendible up to 3-separations (this is called the existence of "a 3-segregation of G and an arrangement of it in Σ compatible with H").

The second change actually causes much more trouble then the first. When we make both of these changes, a new technical problem arises—we have to ensure that the 3-separations of G that we use have H on the proper side, that the baby does not get thrown out with the bath-water. To do so, we introduce "$\mathscr{T}$-local" 3-segregations. The final version of (1.1) is stated in (4.1), and sections 2 and 3 are concerned with definitions.

The proof we give of our main result is not the simplest possible. We have chosen a proof which involves proving and applying a number of fairly complicated lemmas about "crooked transactions"; because we shall need these lemmas in any case in the next paper of the series, and, given these lemmas, deriving our present result is easy.

2. Surfaces

In this paper, a *surface* is a compact connected 2-manifold, possibly with boundary. In fact, the only surfaces with boundary that we shall need to consider are *closed discs* and *closed cylinders*, homeomorphic to

$$\{(x, y) \in \mathbf{R}^2 : x^2 + y^2 \leq 1\}$$

$$\{(x, y) \in \mathbf{R}^2 : 1 \leq x^2 + y^2 \leq 2\}$$

respectively. *Open discs* are defined similarly (but they are not surfaces). If Σ is a surface, its boundary is denoted by $\mathrm{bd}(\Sigma)$; and if $\Lambda \subseteq \Sigma$ is an open disc, its boundary $\mathrm{bd}(\Lambda)$ is defined in the natural way. (We shall use [p. 26] $\Lambda, \Lambda', \Lambda_1$ etc., for open discs, and $\Delta, \Delta', \Delta_1, \ldots$ for closed discs. Σ will always denote a surface.) If $X \subseteq \Sigma$ its closure is denoted by $\bar{X}$. If X is homeomorphic to a circle it is called an *O-arc*; and if X is homeomorphic to $[0, 1]$ it is called a *line*, and its ends are defined in the natural way.

Let Σ be a surface. A *drawing* in Σ is a pair (U, V) where $U \subseteq \Sigma$ is closed, $V \subseteq U$ is finite, and

(i) $U \cap \mathrm{bd}(\Sigma) \subseteq V$

(ii) there are only finitely many arc-wise connected components of $U - V$, called *edges*, and

(iii) each edge e is homeomorphic to a open interval, and $\bar{e} \subseteq V \cup e$, and either $|V \cap \bar{e}| = 1$ and $\bar{e}$ is an *O-arc*, or $|V \cap \bar{e}| = 2$ and $\bar{e}$ is a line with ends the points in $V \cap \bar{e}$.

If $H = (U, V)$ is a drawing in Σ, we define $U(H) = U$ and $V(H) = V$, and $E(H)$ denotes the set of edges of H. We say $v \in V(H)$ is *incident* with $e \in E(H)$ if $v \in \bar{e}$; and in this sense every drawing may be regarded as a graph. We use graph-theoretic terminology for drawings without further explanation. Each component of $\Sigma - U$ is a *region* of H in Σ, and the set of all regions is denoted by $R(H)$. A region r is *incident* with $v \in V(H)$ or $e \in E(H)$ if $v \in \bar{r}$ or $e \subseteq \bar{r}$. A drawing is *2-cell* if every region is an open disc (and consequently $\mathrm{bd}(\Sigma) = \emptyset$). The *atoms* of H are the sets $e(e \in E(H))$, $r(r \in R(H))$ and $\{v\}(v \in V(H))$, and the set of all atoms is denoted by $A(H)$. The atoms thus form a partition of Σ.

If H is a drawing in Σ, and $\Delta \subseteq \Sigma$ is such that either $\bar{e} \subseteq \Delta$ or $e \cap \Delta = \emptyset$ for each $e \in E(H)$, then $(U(H) \cap \Delta, V(H) \cap \Delta)$ is a drawing in Σ which we denote by $H \cap \Delta$.

Let H be a 2-cell drawing in Σ, and let K be another drawing in Σ. We say that K is a *radial drawing* of H if

(i) $V(H) \subseteq V(K)$, and $U(H) \cap U(K) = V(H)$

(ii) K is bipartite, and $(V(H), V(K) - V(H))$ is a 2-colouring of K

(iii) every region of H contains a unique vertex of K, and

(iv) for each $v \in V(H)$, the edges of H and of K incident with v alternate in their cyclic order around v.

(Roughly, K is obtained from $V(H)$ by adding a vertex in each region of H and making it adjacent to every vertex of H incident with that region.)

If K is a radial drawing of H, then to every atom x of H there naturally corresponds an atom x' of K; namely, if $x = \{v\}$ for some $v \in V(H)$ then $x' = x$, if $x \in E(H)$, then $x' \in R(K)$ includes e; and if $x \in R(H)$, then $x' = \{v\}$ for the unique $v \in V(K)$ with $v \in x$. We call x' the *atom of K corresponding to x*.

[p. 27] 3. Tangles

Graphs in this paper are finite, and may have loops or multiple edges. If G_1 is a subgraph of G we write $G_1 \subseteq G$. If $G_1, G_2 \subseteq G$ we denote the graphs

$$(V(G_1) \cup V(G_2), E(G_1) \cup E(G_2))$$

$$(V(G_1) \cap V(G_2), E(G_1) \cap E(G_2))$$

by $G_1 \cup G_2$, $G_1 \cap G_2$ respectively; and G_1, G_2 are disjoint if $V(G_1 \cap G_2) = \emptyset$. A separation of G is a pair (G_1, G_2) of subgraphs of G such that $G_1 \cup G_2 = G$ and $E(G_1 \cap G_2) = \emptyset$; its order is $|V(G_1 \cap G_2)|$. A tangle in G of order θ is a set $\mathscr{T}$ of separations of G, each of order $< \theta$, such that

(i) either $(A, B) \in \mathscr{T}$ or $(B, A) \in \mathscr{T}$, for every separation (A, B) of G of order $< \theta$,

(ii) there do not exist $(A, B), (A', B'), (A'', B'') \in \mathscr{T}$ such that $A \cup A' \cup A'' = G$, and

(iii) $V(A) \neq V(G)$, for all $(A, B) \in \mathscr{T}$.

Tangles were introduced in [4].

Of special interest to us are tangles in drawings. Let H be a 2-cell drawing in Σ, and let $\mathscr{T}$ be a tangle in H of order θ. Let K be a radial drawing of H. We say that $\mathscr{T}$ is respectful if for every circuit C in K of length $< 2\theta$, there is a closed disc $\Delta \subseteq \Sigma$ bounded by $U(C)$ such that $(H \cap \Delta, H \cap \overline{\Sigma - \Delta}) \in \mathscr{T}$. Such a disc Δ is unique, and we write $ins(C) = \Delta$; ins is called the inside function of $\mathscr{T}$. If W is a closed walk in K of length $< 2\theta$, we define $ins(W)$ to be

$$U(K') \cup \bigcup (ins(C) \colon C \text{ is a circuit of } K')$$

where K' is the subgraph of K formed by the vertices and edges of W. Let $x, y \in A(H)$ and let x', y' be the corresponding atoms of K. We define

$$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ k & \text{if } x \neq y \text{ and there is a closed walk } W \text{ of } K \\ & \quad \text{with length } < 2\theta \text{ with } x', y' \subseteq ins(W), \\ & \quad \text{and the shortest such walk has length } 2k \\ \theta & \text{otherwise.} \end{cases}$$

(We remark that every walk in K has even length since K is bipartite). If $v \in V(H)$ we write $d(v, y)$ for $d(\{v\}, y)$ and so on. We call d the *metric of* $\mathscr{T}$. It was introduced, and shown to be a metric, in [5]. The following [p. 28] observation allows us to use the metric without mentioning the corresponding atoms of K.

(3.1) *If W is a closed walk of K with length $< 2\theta$ (so that $ins(W)$ is defined), and $x \in A(H)$, and x' is the corresponding atom of K, then $x' \subseteq ins(W)$ if and only if $x \cap ins(W) \neq \emptyset$.*

The proof is clear. We give the details. Let K' be the subgraph of K formed by the vertices and edges of W, so that $ins(W)$ is the union of $U(K')$ and the discs $ins(C)$ for the circuits C of K'. Since $x \subseteq x'$ unless x is a region of H, and $x' \subseteq x$ if x is a region, the condition $x' \subseteq ins(W)$ implies $x \cap ins(W) \neq \emptyset$. For the converse, let $y \in x \cap ins(W)$. If $x = \{v\}$ with $v \in V(H)$, then $x' = x \subseteq ins(W)$. Suppose that $x \in E(H)$. Then $y \notin U(K)$, since $U(H) \cap U(K) = V(H)$, and so $y \in ins(C)$ for some circuit C of K'. Now x' is a region of K, and so it is connected and disjoint from $U(C)$, which is the boundary of $ins(C)$. Since $y \in x' \cap ins(C)$, it follows that $x' \subseteq ins(C) \subseteq ins(W)$. Finally, suppose that $x \in R(H)$, and let $x' = \{w\}$. The only points of $U(K)$ in x are w and the points of the edges of K incident with w. Hence if $y \in U(K')$, then $w \in V(K') \subseteq ins(W)$. Otherwise $y \in ins(C)$ for some circuit C of K'. If $w \in V(C)$, then $w \in ins(C)$. If not, then x is disjoint from $U(C)$, since C has no edge incident with w. Since x is connected and meets $ins(C)$, it follows that $x \subseteq ins(C)$, and so $w \in ins(C) \subseteq ins(W)$.[F1]

Let H be a 2-cell drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H of order θ, with metric d. Let $z \in A(H)$ and let $\lambda \geq 0$ be an integer. A *λ-zone around z* is an open disc $\Lambda \subseteq \Sigma$ with $z \subseteq \Lambda$, such that $bd(\Lambda)$ is an O-arc and ~~hence~~ $\overline{\Lambda}$ is a closed disc,[F2] $bd(\Lambda) \subseteq U(H)$, $d(z,x) \leq \lambda$ for all $x \in A(H)$ with $x \subseteq \overline{\Lambda}$, and if $z \in E(H)$ then $\lambda \geq 2$. A *λ-zone (for H)* is a subset of Σ which is a λ-zone around z for some $z \in A(H)$. Thus, if Λ is a λ-zone then $H \cap bd(\Lambda)$ is a circuit.

Let Λ be a λ-zone around z and let $H' = H \cap (\Sigma - \Lambda)$. Since $H \cap bd(\Lambda)$ is a circuit it follows that Λ is a region of H', and every other region of H' is a region of H. Consequently H' is a 2-cell drawing in Σ. We say that $\mathscr{T}'$ is a *tangle of order $\theta - 4\lambda - 2$ obtained by clearing* Λ if $\mathscr{T}'$ is a tangle in H' of order $\theta - 4\lambda - 2$, and

(i) $\mathscr{T}'$ is respectful, and has metric d' say

(ii) $(A \cap H', B \cap H') \in \mathscr{T}'$ for all $(A, B) \in \mathscr{T}$ of order $< \theta - 4\lambda - 2$, and

(iii) if $x, y \in A(H)$ and $x', y' \in A(H')$ with $x \subseteq x'$ and $y \subseteq y'$, then $d(x,y) \geq d'(x',y') \geq d(x,y) - 4\lambda - 2$.

[F1] **Missing proof.** The paper says only that the proof of (3.1) is clear. The argument above is the one used in the formalization.

[F2] **Statement mistake.** That $\overline{\Lambda}$ is a closed disc does not follow from the other conditions. In the projective plane, seen as a closed disc with opposite boundary points identified, the image of the open disc is an open disc Λ whose frontier is an O-arc, but $\overline{\Lambda}$ is the whole projective plane. So the condition is made part of the definition. Every λ-zone built in Section 9 has this property, so the results about λ-zones hold as printed.

It was shown in [6, theorem (7.10)] that

(3.2) *If $\theta \geq 4\lambda + 3$ then there is a unique tangle of order $\theta - 4\lambda - 2$ obtained by clearing Λ.*

In general, if $\mathscr{T}$ is a tangle of order θ in a graph H then $X \subseteq V(H)$ is *free with respect to* $\mathscr{T}$ if $|X| \leq \theta$ and there is no $(A, B) \in \mathscr{T}$ of order $< |X|$ with $X \subseteq V(A)$. It is easy to see and was shown in [4, theorem (12.2)] that any subset of a free set is free.

If H is a subgraph of G, an *H-path* is a path of G with distinct ends both in $V(H)$ and with no other vertex or edge in H. Let H be a subgraph of G, and let H be a 2-cell drawing in Σ. Let $\mathscr{T}$ be a respectful tangle in H. A region r of H is an *eye of H (in G) with respect to* $\mathscr{T}$ if

(i) there is a circuit C of H with $U(C) = bd(r)$

(ii) there are distinct vertices a, b, c, d occurring in $V(C)$ in that order, and $\{a, b, c, d\}$ is free with respect to $\mathscr{T}$, and

[p. 29] (iii) there are two disjoint H-paths in G with ends a, c and b, d respectively.

4. Statement Of The Theorem

To state our main result precisely we need several further definitions. Let H be a drawing in a surface Σ. If $F \subseteq \Sigma$ is an O-arc with $F \cap U(H) \subseteq V(H)$ and $|F \cap V(H)| \leq 2$, a *dial for F (or for F, H in case of ambiguity)* is a closed disc $\Delta \subseteq \Sigma$ with $bd(\Delta) = F$, such that either

(i) $\Delta \cap U(H) \subseteq F$, or

(ii) $|F \cap V(H)| = 2$ and $H \cap \Delta$ is a path joining the two vertices in $F \cap V(H)$.

We say that H is *rigid (in Σ)* if Σ includes a dial for every O-arc $F \subseteq \Sigma$ with $F \cap U(H) \subseteq V(H)$ and $|F \cap V(H)| \leq 2$.

Let Ω be a cyclic permutation of the elements of some set. We denote this set by $\overline{\Omega}$. If V is a set and $\overline{\Omega} \subseteq V$, Ω is said to be a *cyclic permutation in V*. If F is an O-arc, and $X \subseteq F$ is finite, then both orientations of F induce cyclic permutations of X, and we call these the *natural orders of X from F*.

A *society* is a pair (G, Ω), where G is a graph and Ω is a cyclic permutation in $V(G)$. A *segregation* of a graph G is a set $\mathscr{S}$ of societies such that

(i) $A \subseteq G$ for every $(A, \Omega) \in \mathscr{S}$, and $\bigcup(A\colon (A, \Omega) \in \mathscr{S}) = G$, and

(ii) $V(A \cap A') \subseteq \overline{\Omega} \cap \overline{\Omega}'$ and $E(A \cap A') = \emptyset$ for all distinct $(A, \Omega), (A', \Omega') \in \mathscr{S}$.

We write $V(\mathscr{S}) = \bigcup(\overline{\Omega}\colon (A,\Omega) \in \mathscr{S})$. A segregation $\mathscr{S}$ is a *3-segregation* if $|\overline{\Omega}| \leq 3$ for all $(A,\Omega) \in \mathscr{S}$. Actually, in this paper all the segregations we consider will be 3-segregations.

Let $\mathscr{S} = \{(A_1,\Omega_1),\ldots,(A_k,\Omega_k)\}$ be a segregation of G, and let Σ be a surface. An *arrangement of $\mathscr{S}$ in* Σ is a function α with domain $\mathscr{S} \cup V(\mathscr{S})$, such that (writing $\alpha(A,\Omega)$ for $\alpha((A,\Omega))$):

(i) $1 \leq i \leq k$, $\alpha(A_i,\Omega_i)$ is a closed disc $\Delta_i \subseteq \Sigma$, and $\alpha(x) \in bd(\Delta_i)$ for each $x \in \overline{\Omega}_i$

(ii) for $1 \leq i < j \leq k$, if $x \in \Delta_i \cap \Delta_j$ then $x = \alpha(v)$ for some $v \in \overline{\Omega}_i \cap \overline{\Omega}_j$; and for $1 \leq i \leq k$, if $x \in \Delta_i \cap bd(\Sigma)$ then $x = \alpha(v)$ for some $v \in \overline{\Omega}_i$,

(iii) for distinct $x, y \in V(\mathscr{S})$, $\alpha(x) \neq \alpha(y)$, and

(iv) for $1 \leq i \leq k$, Ω_i is mapped by α to a natural order of $\alpha(\Omega_i)$ from $bd(\Delta_i)$.

If $\mathscr{S}$ is a 3-segregation then condition (iv) here is trivial.

[p. 30] Now let H be a subgraph of G, and let H be a 2-cell drawing in Σ. Let α be an arrangement in Σ of a segregation $\mathscr{S}$ of G. We say that α is *compatible with H* if

(i) $\alpha(v) = v$ for all $v \in V(H) \cap V(\mathscr{S})$, and

(ii) for each $(A,\Omega) \in \mathscr{S}$, $U(H) \cap bd(\alpha(A,\Omega)) = V(H) \cap \overline{\Omega}$, and $H \cap \alpha(A,\Omega) = H \cap A$.

If H is a subgraph of G and $\mathscr{S}$ is a segregation of G, and $\mathscr{T}$ is a tangle in H, we say that $\mathscr{S}$ is *$\mathscr{T}$-local* if $(A \cap H, B \cap H) \in \mathscr{T}$ for all $(A,\Omega) \in \mathscr{S}$, where B is the unique subgraph of G such that (A,B) is a separation of G and $V(A \cap B) = \overline{\Omega}$.

Finally we can state our main theorem.

(4.1) *Let H be a 2-cell rigid drawing in a surface Σ, and let $\mathscr{T}$ be a respectful tangle in H of order $\theta \geq 209$ with metric d. Let H be a subgraph of G, and suppose that*

(i) *$d(u,v) \leq 3$ for every H-path in G with ends u, v, and*

(ii) *if $\mathscr{T}'$ is the tangle of order $\theta - 158$ obtained by clearing some 39-zone Λ such that $H \cap (\Sigma - \Lambda)$ is a rigid drawing in Σ, then Λ is not an eye of $H \cap (\Sigma - \Lambda)$ in G with respect to $\mathscr{T}'$.*

Then there is a $\mathscr{T}$-local 3-segregation of G and an arrangement of it in Σ compatible with H.

The idea of the proof is not very difficult. A bridge of H in G is a subgraph of G, consisting of either an edge of G not in $E(H)$ but with both ends in $V(H)$, or a component of $G \setminus V(H)$ and all edges between it and $V(H)$ (and their ends). We choose a bridge J of H in G, and choose

$u \in V(J) \cap V(H)$. By hypothesis, $d(u,v) \leq 3$ for all other $v \in V(J) \cap V(H)$. By induction on the size of G, the result is true for the graph G' obtained from G by deleting the edges of J and the vertices of J not in $V(H)$. Choose a 3-segregation of G' and an arrangement of it satisfying the theorem. Now all the attachments of J are close to u, and so we can find a small open disc Λ around u containing all the attachments of J. If we can rearrange the 3-segregation within Λ to accommodate J as well, then the induction step is complete. (There is a problem about keeping it compatible with H, but that is easy because H is rigid.) If we cannot rearrange it in this way, then this part of G' together with J contains a "cross", two crossing paths with their ends on the boundary of the disc, by a theorem of an earlier paper, and we would like to use this cross to contradict (ii). For this reason, we need to insist that the four end-vertices of the cross form a free set with respect to the tangle obtained by clearing Λ. To do so, we shall arrange that $bd(\Lambda) = U(C)$ for some circuit C in the middle of a large sequence of [p. 31] disjoint circuits, forming the boundaries of a nested sequence of discs all containing u, and that there is a large number of disjoint paths between the inner and outer circuits. In these circumstances a theorem of [3] can be applied to arrange that the end-vertices of the cross form a free set. There is only one other serious problem; it is natural to choose Λ so that its boundary is a circuit of H, because then hypothesis (ii) provides information about Λ. But if we do so, then certain members of the 3-segregation of G' may be arranged partly inside and partly outside Λ, and this causes confusion when we try to rearrange the part of the 3-segregation within Λ. To overcome this we use two discs, one bounded by a circuit of H, and the other, slightly larger, chosen so that no member of the 3-segregation is arranged partly inside and partly outside it.

The main part of the proof in this paper is concerned with proving the existence of the desired two discs. These lemmas are expressed in greater generality than is needed here, because the more general form (involving "crooked transactions") will be needed in the next paper of the series.

5. Circuits in Cylinders

We begin with some results about planar graphs. First, we need the following. A set $\mathscr{R}$ of regions of a drawing H is *connected* if for all $r, r' \in \mathscr{R}$ there is a sequence $r = r_1, r_2, \ldots, r_k = r'$ of members of $\mathscr{R}$ such that for $1 \leq i < k$, there is an edge of H incident with r_i and with r_{i+1}.

(5.1) *Let H be a drawing in a surface Σ, and let $\Delta \subseteq \Sigma$ be a closed disc. Let $\mathscr{R}$ be a non-empty connected set of regions of H, such that $r \subseteq \Delta - bd(\Delta)$*

for all $r \in \mathscr{R}$. Then there is a circuit C of H with $U(C) \subseteq \Delta$, bounding a disc in Δ including each member of $\mathscr{R}$, and such that every edge of C is incident with a member of $\mathscr{R}$.

This result is closely related to the elementary result that the minimal edge-cutsets of the dual graph of a planar graph correspond to the circuits of the graph, and may be proved similarly. ~~We omit further details.~~

We give the details, using some results of [5].

Proof of (5.1). We may assume that $\mathrm{bd}(\Sigma) = \emptyset$. For otherwise, let Σ' be the surface obtained from Σ by pasting a closed disc onto each component of $\mathrm{bd}(\Sigma)$. Then H is a drawing in Σ', and Δ is a closed disc in Σ'. Let $r \in \mathscr{R}$. Then r is connected, and it is open in Σ', since it is open in Σ and disjoint from $\mathrm{bd}(\Sigma)$ (because $r \subseteq \Delta - \mathrm{bd}(\Delta)$). Moreover every point $x \in \bar{r} - r$ is in $U(H)$, for otherwise the region of H in Σ that contains x is an open set meeting r, and so equals r, which is impossible since $x \notin r$. Hence r is also a region of H in Σ'. The incidences between the edges of H and the members of $\mathscr{R}$ are the same in Σ and in Σ', and so (5.1) for Σ' implies (5.1) for Σ.

Thus we assume that $\mathrm{bd}(\Sigma) = \emptyset$. Let N be the subdrawing of H formed by the vertices and edges of H that are incident both with a member of $\mathscr{R}$ and with a region of H not in $\mathscr{R}$. (So $N = N_H(\mathscr{R})$ in the notation of [5].) Every vertex and every edge of N is a subset of $\bar{r}$ for some $r \in \mathscr{R}$, and $\bar{r} \subseteq \Delta$ since Δ is closed. Hence $U(N) \subseteq \Delta$. For each circuit C of N, $U(C)$ is an O-arc in the closed disc Δ, and so it bounds a unique closed disc $\mathrm{ins}(C) \subseteq \Delta$. Let $\theta = |E(N)|/2 + 1$. Every circuit of N has length at most $|E(N)| < 2\theta$. We claim that ins is a slope in N of order θ, in the sense of [5]. For the first slope axiom, let C_1, C_2 be circuits of N with $U(C_1) \subseteq \mathrm{ins}(C_2)$. Then $U(C_1)$ bounds a closed disc included in $\mathrm{ins}(C_2)$, and this disc is $\mathrm{ins}(C_1)$ by uniqueness, since $\mathrm{ins}(C_2) \subseteq \Delta$. For the second, let P_0, P_1, P_2 be as in that axiom. If $U(P_0) \subseteq \mathrm{ins}(P_1 \cup P_2)$, then $\mathrm{ins}(P_1 \cup P_2)$ includes $U(P_2 \cup P_0)$ and $U(P_0 \cup P_1)$, and hence includes $\mathrm{ins}(P_2 \cup P_0)$ and $\mathrm{ins}(P_0 \cup P_1)$, by the first axiom. The same holds if $U(P_1) \subseteq \mathrm{ins}(P_2 \cup P_0)$ or $U(P_2) \subseteq \mathrm{ins}(P_0 \cup P_1)$, and so we may assume that none of these three inclusions holds. Since $U(P_0) \not\subseteq \mathrm{ins}(P_1 \cup P_2)$ and $U(P_1) \not\subseteq \mathrm{ins}(P_2 \cup P_0)$, the closed discs $\mathrm{ins}(P_1 \cup P_2)$ and $\mathrm{ins}(P_2 \cup P_0)$ meet only in $U(P_2)$, and their union is a closed disc bounded by $U(P_0 \cup P_1)$. This union is a subset of Δ, and so it equals $\mathrm{ins}(P_0 \cup P_1)$ by uniqueness. But it includes $U(P_2)$, a contradiction. This proves our claim. We apply the results of [5, Section 4] with $\Gamma = \Gamma' = N$. Since every circuit of N has length $< 2\theta$, N is confined.

(1) *All the members of $\mathscr{R}$ are included in the same region of N.*

For let $r_1, r_2 \in \mathscr{R}$ be distinct, and both incident with an edge e of H. Since e is incident with at most two regions of H, it is incident with no

region outside $\mathscr{R}$, and so $e \notin E(N)$. Choose $x \in e$. Then $x \notin U(N)$, since $e \notin E(N)$, and so x belongs to a region r' of N. Now $x \in \bar{r}_1 \cap \bar{r}_2$, and so r' meets both r_1 and r_2. Since r_1, r_2 are connected and disjoint from $U(N)$, they are both included in r'. Since $\mathscr{R}$ is connected, (1) follows.

Let r' be the region of N that includes all the members of $\mathscr{R}$. Let $\mathrm{ins}(N)$ be the union of $U(N)$ and of the discs $\mathrm{ins}(C)$ for all circuits C of N. By [5, (4.4)], $r_0 = \Sigma - \mathrm{ins}(N)$ is a region of N. Since $\mathrm{ins}(N) \subseteq \Delta$, it follows that $\Sigma - \Delta \subseteq r_0$. Now $\Sigma - \Delta$ is open, and it is non-empty, for otherwise $\Delta = \Sigma$, which is impossible since $\mathrm{bd}(\Delta) \neq \emptyset$ and $\mathrm{bd}(\Sigma) = \emptyset$. Since $U(H)$ includes no non-empty open subset of Σ, there is a point $y \in \Sigma - \Delta$ with $y \notin U(H)$. Let r_2 be the region of H that contains y. Then $r_2 \notin \mathscr{R}$, since $y \notin \Delta$, and $r_2 \subseteq r_0$, since r_2 is connected, disjoint from $U(N)$, and contains $y \in r_0$.

(2) $r' \neq r_0$.

For suppose that $r' = r_0$, and choose $r_1 \in \mathscr{R}$. Then $r_1, r_2 \subseteq r_0$, and so by [5, (5.1)], applied with $\Gamma = H$, $\Gamma' = N$ and $Z = \mathscr{R}$ (its proof does not use the slope), there are regions $r_3 \in \mathscr{R}$ and $r_4 \notin \mathscr{R}$ of H, both incident with some edge e of H with $e \notin E(N)$. But then $e \in E(N)$, from the definition of N, a contradiction. This proves (2).

Now N does not capture r_0, since $y \in r_0$ and $y \notin \mathrm{ins}(N)$. In the proof of [5, (4.4)] it is shown that a confined subdrawing captures at least one of any two distinct regions of it. By (2), it follows that N captures r', that is, there is a circuit C of N with $r' \subseteq \mathrm{ins}(C)$. Then $U(C) \subseteq U(N) \subseteq \Delta$, and $U(C)$ bounds the disc $\mathrm{ins}(C) \subseteq \Delta$, which includes each member of $\mathscr{R}$ by (1). Every edge of C is an edge of N, and so is incident with a member of $\mathscr{R}$. This proves (5.1). □[F3]

(5.2) *Let H be a drawing in Σ, and let $\Delta \subseteq \Sigma$ be a closed disc. Let $Z \subseteq \Delta - bd(\Delta)$ be non-empty and arc-wise connected, and such that $r \subseteq \Delta - bd(\Delta)$ for every region r of H with $\bar{r} \cap Z \neq \emptyset$. Then there is a circuit C of H with $U(C) \subseteq \Delta$, bounding an open disc in Δ including Z, such that every edge of C is incident with a region r of H satisfying $\bar{r} \cap Z \neq \emptyset$.*

Proof. Let $\mathscr{R}$ be the set of all regions r of H such that $\bar{r} \cap Z \neq \emptyset$. By (5.1) it suffices to show that $\mathscr{R}$ is connected, and this follows easily from the connectedness of Z. We give the details. Since $U(H)$ includes no non-empty open subset of Σ, every point of Σ is in the closure of some region of H. In particular $\mathscr{R} \neq \emptyset$, since $Z \neq \emptyset$. To show that $\mathscr{R}$ is connected, choose $r_0 \in \mathscr{R}$, let $\mathscr{R}_1$ be the set of all $r \in \mathscr{R}$ for which there is a sequence from r_0 to r

[F3] **Missing proof.** The paper omits the proof of (5.1). Apart from the reduction in the first paragraph to a surface without boundary, the argument above is the one used in the formalization.

as in the definition of connectedness, and let $\mathscr{R}_2 = \mathscr{R} - \mathscr{R}_1$. Suppose that $\mathscr{R}_2 \neq \emptyset$. Let K_1, K_2 and K_3 be the unions of the closures of the members of $\mathscr{R}_1$, of the members of $\mathscr{R}_2$, and of the regions of H not in $\mathscr{R}$, respectively. These sets are closed, since H has only finitely many regions. Every point of Σ not in K_3 is in $K_1 \cup K_2$, and $Z \cap K_3 = \emptyset$ by the definition of $\mathscr{R}$. We claim that $K_1 \cap K_2 \subseteq V(H)$. For let $x \in \bar{r}_1 \cap \bar{r}_2$, where $r_1 \in \mathscr{R}_1$ and $r_2 \in \mathscr{R}_2$. If $x \notin U(H)$, then the region of H that contains x is an open set meeting r_1 and r_2, and so $r_1 = r_2$, which is impossible. If x belongs to an edge e of H, then e is incident with r_1 and r_2, and so $r_2 \in \mathscr{R}_1$, which is impossible. This proves our claim.

Next we show that $Z \cap K_1 \cap K_2 = \emptyset$. For suppose that $v \in Z \cap K_1 \cap K_2$. Then $v \in V(H)$ by our claim, and $v \notin K_3$. Since $v \in \Delta - \mathrm{bd}(\Delta)$, v is not in $\mathrm{bd}(\Sigma)$, and so there is an open disc $W \subseteq \Sigma$ containing v and disjoint from K_3 and from $V(H) - \{v\}$. Then $W - \{v\}$ is connected. Each point of $W - \{v\}$ is in $K_1 \cup K_2$, since it is not in K_3, and it is not in both K_1 and K_2, by our claim, since it is not in $V(H)$. Since K_1, K_2 are closed, it follows that $W - \{v\}$ is disjoint from one of K_1, K_2, say from K_i. But $v \in K_i$, and so $v \in \bar{r}$ for some region r with $\bar{r} \subseteq K_i$. Then r meets W, and $r \cap W \subseteq W - \{v\}$ since $v \in U(H)$. Hence $W - \{v\}$ meets K_i, a contradiction. Thus $Z \cap K_1 \cap K_2 = \emptyset$.

Now $Z \subseteq K_1 \cup K_2$, since $Z \cap K_3 = \emptyset$, and Z meets both K_1 and K_2, since $\mathscr{R}_1$ and $\mathscr{R}_2$ are non-empty subsets of $\mathscr{R}$. Hence Z is the union of the disjoint non-empty sets $Z \cap K_1$ and $Z \cap K_2$, which are closed in Z. This is contrary to the connectedness of Z, since Z is arc-wise connected. Thus $\mathscr{R}_2 = \emptyset$, and $\mathscr{R}$ is connected.

By (5.1), there is a circuit C of H with $U(C) \subseteq \Delta$, bounding a closed disc $D \subseteq \Delta$ that includes each member of $\mathscr{R}$, such that every edge of C is incident with a member of $\mathscr{R}$. It remains to show that $Z \subseteq D - U(C)$, the open disc bounded by $U(C)$. The set $\Sigma - K_3$ is open, and it includes Z. Moreover $\Sigma - K_3 \subseteq D$. For otherwise $(\Sigma - K_3) - D$ is a non-empty open set, and so it contains a point $y \notin U(H)$. Let r be the region of H that contains y. Then $r \notin \mathscr{R}$, since $y \notin D$ and every member of $\mathscr{R}$ is a subset of D, and so $y \in K_3$, a contradiction. Hence Z is included in the interior of D, which is $D - U(C)$. This proves (5.2).[F4] □

[p. 32] If H is a drawing in Σ, a *chain* is a sequence $v_0, r_1, \ldots, r_k, v_k$ such that $r_1, \ldots, r_k$ are regions of H and for $1 \leq i \leq k$, $\{v_{i-1}, v_i\} \subseteq \bar{r}_i$. (We do

[F4] **Missing proof.** The paper says only that the connectedness of $\mathscr{R}$ follows easily from that of Z, and does not show that the disc given by (5.1) has Z in its interior. The argument above is the one used in the formalization.

not insist that $v_0, \ldots, v_k$ are vertices.) The chain has *length* k, and v_0, v_k are its *ends*. If $X, Y \subseteq \Sigma$ are non-empty, we define the *regional distance* between X and Y to be the minimum length of a chain with one end in X and the other in Y. From (5.2) we deduce

(5.3) *Let H be a drawing in Σ, and let $\Delta \subseteq \Sigma$ be a closed disc. Let $Z \subseteq \Delta - bd(\Delta)$ be non-empty and arc-wise connected, such that the regional distance between Z and $\Sigma - \Delta$ is $\geq k \geq 2$. Then there is a circuit C of H with $U(C) \subseteq \Delta$, bounding an open disc Λ including Z, such that the regional distance between $\bar{\Lambda}$ and $\Sigma - \Delta$ is $\geq k - 1$.*

Proof. If r is a region of H and $\bar{r} \cap Z \neq \emptyset$ then $r \subseteq \Delta - \mathrm{bd}(\Delta)$; for otherwise $\bar{r} \cap (\Sigma - \Delta) \neq \emptyset$ and the regional distance between Z and $\Sigma - \Delta$ is ≤ 1, a contradiction. Hence (5.2) may be applied. Let C be as in (5.2), with $U(C)$ bounding an open disc $\Lambda \subseteq \Delta$. We claim that the regional distance between $\bar{\Lambda}$ and $\Sigma - \Delta$ is $\geq k - 1$. For let $v_0, r_1, v_1, \ldots, r_{k'}, v_{k'}$ be a chain with $v_0 \in \bar{\Lambda}$ and $v_{k'} \in \Sigma - \Delta$, and with k' minimum. Now $k' \geq 1$ since $\bar{\Lambda} \subseteq \Delta$, and $v_1 \notin \bar{\Lambda}$ since k' is minimum. Hence $r_1 \not\subseteq \Lambda$, and so $r_1 \cap \Lambda = \emptyset$, and $v_0 \in V(C)$. Let e be an edge of C incident with v_0. From the choice of C there is a region r of H incident with e satisfying $\bar{r} \cap Z \neq \emptyset$. Choose $v \in \bar{r} \cap Z$; then from the chain

$$v, r, v_0, r_1, v_1, \ldots, r_{k'}, v_{k'}$$

and the hypothesis that the regional distance between Z and $\Sigma - \Delta$ is at least k, we deduce that $k' \geq k - 1$, as required. □

By repeated application of (5.3), we deduce

(5.4) *Let H, Σ, Δ, Z, k be as in (5.3). Then there are mutually disjoint circuits $C_1, \ldots, C_{k-1}$ of H with $U(C_1), \ldots, U(C_{k-1}) \subseteq \Delta$, all bounding open discs including Z.*

A similar proof ~~(we omit the details)~~ yields

(5.5) *Let H be a drawing in Σ, and let $\Delta \subseteq \Sigma$ be a closed disc. Let $Z \subseteq \Delta - bd(\Delta)$ be non-empty and arc-wise connected, such that the regional distance between Z and $bd(\Delta)$ is $\geq k \geq 1$. Then there are $k - 1$ mutually disjoint circuits $C_1, \ldots, C_{k-1}$ of H with $U(C_1), \ldots, U(C_{k-1}) \subseteq \Delta - bd(\Delta)$, all bounding open discs including Z.*

Proof of (5.5). We first show the following analogue of (5.3).

(1) *If $k \geq 2$, there is a circuit C of H with $U(C) \subseteq \Delta - bd(\Delta)$, bounding an open disc Λ including Z, such that $\bar{\Lambda} \subseteq \Delta - bd(\Delta)$ and the regional distance between $\bar{\Lambda}$ and $bd(\Delta)$ is $\geq k - 1$.*

For let r be a region of H with $\bar{r} \cap Z \neq \emptyset$. Then $\bar{r} \cap \mathrm{bd}(\Delta) = \emptyset$, for otherwise the regional distance between Z and $\mathrm{bd}(\Delta)$ is ≤ 1. In particular

r is disjoint from $\mathrm{bd}(\Delta)$. Moreover r meets $\Delta - \mathrm{bd}(\Delta)$, since $\bar{r}$ meets $Z \subseteq \Delta - \mathrm{bd}(\Delta)$ and $\Delta - \mathrm{bd}(\Delta)$ is open. Since r is connected, it follows that $r \subseteq \Delta$, and so $\bar{r} \subseteq \Delta - \mathrm{bd}(\Delta)$. In particular (5.2) may be applied. Let C be as in (5.2), bounding an open disc $\Lambda \subseteq \Delta$ including Z. Every edge e of C is incident with a region r of H with $\bar{r} \cap Z \neq \emptyset$, and so $\bar{e} \subseteq \bar{r} \subseteq \Delta - \mathrm{bd}(\Delta)$. Since every vertex of C is an end of an edge of C, it follows that $U(C) \subseteq \Delta - \mathrm{bd}(\Delta)$. Since Λ is open and $\Lambda \subseteq \Delta$, we have $\Lambda \cap \mathrm{bd}(\Delta) = \emptyset$, and so $\bar{\Lambda} = \Lambda \cup U(C) \subseteq \Delta - \mathrm{bd}(\Delta)$. Now let $v_0, r_1, v_1, \ldots, r_{k'}, v_{k'}$ be a chain with $v_0 \in \bar{\Lambda}$ and $v_{k'} \in \mathrm{bd}(\Delta)$, with k' minimum. Then $k' \geq 1$, since $\bar{\Lambda} \cap \mathrm{bd}(\Delta) = \emptyset$. Also $v_1 \notin \bar{\Lambda}$, by the minimality of k' if $k' \geq 2$, and since $v_1 \in \mathrm{bd}(\Delta)$ if $k' = 1$. Hence $r_1 \not\subseteq \Lambda$, for otherwise $v_1 \in \bar{r}_1 \subseteq \bar{\Lambda}$. Since r_1 is connected and disjoint from $U(C)$, it follows that $r_1 \cap \Lambda = \emptyset$, and so $\bar{r}_1 \cap \Lambda = \emptyset$, since Λ is open. Thus $v_0 \notin \Lambda$, since $v_0 \in \bar{r}_1$, and therefore $v_0 \in U(C)$. Let e be an edge of C with $v_0 \in \bar{e}$, and let r be a region of H incident with e such that $\bar{r} \cap Z \neq \emptyset$. Choose $v \in \bar{r} \cap Z$. Then $v_0 \in \bar{e} \subseteq \bar{r}$, and so

$$v, r, v_0, r_1, v_1, \ldots, r_{k'}, v_{k'}$$

is a chain of length $k' + 1$ between Z and $\mathrm{bd}(\Delta)$. Hence $k' + 1 \geq k$. This proves (1).

We prove (5.5) by induction on k. If $k = 1$ there is nothing to prove, and so we assume that $k \geq 2$. Let C and Λ be as in (1). Then $\bar{\Lambda}$ is a closed disc, and so it is non-empty and arc-wise connected. Moreover $\bar{\Lambda} \subseteq \Delta - \mathrm{bd}(\Delta)$, and the regional distance between $\bar{\Lambda}$ and $\mathrm{bd}(\Delta)$ is $\geq k - 1 \geq 1$. From our inductive hypothesis, applied with $\bar{\Lambda}$ in place of Z, there are $k-2$ mutually disjoint circuits $C_2, \ldots, C_{k-1}$ of H with $U(C_2), \ldots, U(C_{k-1}) \subseteq \Delta - \mathrm{bd}(\Delta)$, bounding open discs $\Lambda_2, \ldots, \Lambda_{k-1}$ that include $\bar{\Lambda}$, and hence include Z. For $2 \leq i \leq k-1$, $U(C) \subseteq \bar{\Lambda} \subseteq \Lambda_i$ and $U(C_i) \cap \Lambda_i = \emptyset$, and so $U(C) \cap U(C_i) = \emptyset$. Setting $C_1 = C$, the circuits $C_1, \ldots, C_{k-1}$ satisfy (5.5). $\square$[F5]

[p. 33] ## 6. Rigidity

In this section we prove an assortment of lemmas about rigidity.

(6.1) *Let H be a rigid drawing in a surface Σ, and let $\Sigma' \subseteq \Sigma$ be a surface, such that $bd(\Sigma') \cap U(H) \subseteq V(H)$. Suppose that for every closed disc $\Delta' \subseteq \Sigma$ with $\Delta' \cap \Sigma' = bd(\Delta') \subseteq bd(\Sigma')$ there is a circuit C of H with $U(C) \subseteq \Delta'$. Then $H \cap \Sigma'$ is a rigid drawing in Σ'.*

[F5] **Missing proof.** The paper omits the proof of (5.5). The argument above is the one used in the formalization.

Proof. Let $F \subseteq \Sigma'$ be an O-arc with $F \cap U(H') \subseteq V(H')$ and $|F \cap V(H')| \leq 2$, where $H' = H \cap \Sigma'$. Since $F \cap U(H') = F \cap U(H)$ and H is rigid in Σ, there is a dial $\Delta \subseteq \Sigma$ for F. We claim that $\Delta \subseteq \Sigma'$. For if not then since $\mathrm{bd}(\Delta) \subseteq \Sigma'$ it follows that there is a closed disc $\Delta' \subseteq \Delta$ with $\Delta' \cap \Sigma' = \mathrm{bd}(\Delta') \subseteq \mathrm{bd}(\Sigma')$, and hence there is a circuit C of H with $U(C) \subseteq \Delta' \subseteq \Delta$, a contradiction. Hence $\Delta \subseteq \Sigma'$, and so H' is rigid in Σ', as required. □

A subset $X \subseteq \Sigma$ is *simply-connected* if X is arc-wise connected and every O-arc $F \subseteq X$ bounds a closed disc $\Delta \subseteq X$.

(6.2) *Let H be a rigid drawing in a surface Σ. Let $v \in V(H)$, and let the regions of H incident with v be $r_1, \ldots, r_k$, enumerated in the cyclic order of their occurrence around v. Then $r_1, \ldots, r_k$ are all distinct. Consequently $\bar{r}$ is simply-connected for every region r of H.*

Proof. Suppose that $r_i = r_j$ where $1 \leq i < j \leq k$. Then there is an O-arc $F \subseteq r_i \cup \{v\}$ such that no closed disc $\Delta \subseteq \Sigma$ bounded by F has $E(H \cap \Delta) = \emptyset$, contrary to the rigidity of H. Thus $r_1, \ldots, r_k$ are all distinct. Hence no vertex "occurs twice" on the boundary of one region r, and hence to show that $\bar{r}$ is simply-connected it suffices to show that r is simply-connected. Let $F \subseteq r$ be an O-arc. Then $F \cap U(H) = \emptyset$, and so there is a closed disc $\Delta \subseteq \Sigma$ bounded by F with $\Delta \cap U(H) = \emptyset$, since H is rigid. Then $\Delta \subseteq r$; and so r is simply-connected, as required. □

(6.3) *Let H be a rigid drawing in a surface Σ, and let $\Lambda \subseteq \Sigma$ be an open disc with $bd(\Lambda) \subseteq U(H)$ such that $bd(\Lambda)$ is an O-arc. Suppose that $\bar{r} \cap \bar{\Lambda}$ is null or connected for every region r of H with $r \cap \Lambda = \emptyset$. Then $H \cap (\Sigma - \Lambda)$ is a rigid drawing in Σ.*

Proof. Let $H' = H \cap (\Sigma - \Lambda)$, and C be the circuit $H \cap \mathrm{bd}(\Lambda)$. Let $F \subseteq \Sigma$ be an O-arc with $F \cap U(H') \subseteq V(H')$ and $|F \cap V(H')| \leq 2$. We must prove that there is a dial for F, H'.

By (6.1), H' is a rigid drawing in $\Sigma - \Lambda$, and so we may assume that $F \not\subseteq \Sigma - \Lambda$, that is, $F \cap \Lambda \neq \emptyset$. If $F \subseteq \bar{\Lambda}$ then the disc in $\bar{\Lambda}$ bounded by F is a dial for F, H' and so we may assume that $F \not\subseteq \bar{\Lambda}$. Since $|F \cap V(H')| \leq 2$ it follows that $|F \cap V(H')| = 2$, $F \cap V(H') = \{v_1, v_2\}$, say, [p. 34] and $v_1, v_2 \in V(C)$, and there is a region $r \neq \Lambda$ of H' incident with v_1 and v_2, and $F_1 \subseteq \Lambda$ and $F_2 \subseteq r$, where F_1, F_2 are the two components of $F - \{v_1, v_2\}$. From the hypothesis, $\bar{r} \cap \bar{\Lambda}$ is connected, and so there is a line $L \subseteq \bar{r} \cap \bar{\Lambda}$ with ends v_1, v_2. Let $\Delta_1 \subseteq \bar{\Lambda}$ be the closed disc bounded by the O-arc $F_1 \cup L$; and let $\Delta_2 \subseteq \bar{r}$ be the closed disc bounded by $F_2 \cup L$. (These exist because $\bar{\Lambda}$ is a closed disc, and $\bar{r}$ is simply-connected by (6.2).) Then $\Delta = \Delta_1 \cup \Delta_2$ is a closed disc bounded by F. Moreover, $H' \cap \Delta_1 = H' \cap L$, and $H' \cap \Delta_2 = H' \cap L$

by (6.2). Hence $H \cap \Delta = H \cap L$, as required. □

Two drawings are *isomorphic* if they are isomorphic as graphs.

(6.4) *Let Δ be a closed disc, let H_1, H_2 be drawings in Δ with $V(H_1) \cap bd(\Delta) = V(H_2) \cap bd(\Delta)$, and let $\phi\colon H_1 \to H_2$ be an isomorphism such that $\phi(v) = v$ for all $v \in V(H_1) \cap bd(\Delta)$. If H_1 is rigid in Δ then there is a homeomorphism $\psi\colon \Delta \to \Delta$ extending ϕ.*

Proof. Since H_1 is rigid, no vertex of H_1 in $\Delta - \mathrm{bd}(\Delta)$ has valency ≤ 1; and we may assume that no vertex of H_1 in $\Delta - \mathrm{bd}(\Delta)$ has valency ≤ 2, by suppressing such vertices. We may assume that $E(H_1) \neq \emptyset$ and H_1 is not a path, and so $|V(H_1) \cap \mathrm{bd}(\Delta)| \geq 3$ since H_1 is rigid. Let $u, v \in V(H_1) \cap \mathrm{bd}(\Delta)$ be consecutive in $\mathrm{bd}(\Delta)$, and let r_1 be the region of H_1 in Δ including the part of $\mathrm{bd}(\Delta)$ between u and v. Define r_2 similarly, from H_2. If u, v are not adjacent in H_1 we may add a new edge to H_1 and to H_2 joining u, v drawn within the regions r_1, r_2 respectively; and the first enlarged drawing is still rigid, and still isomorphic to the second, and if the result holds for this pair then it holds for H_1, H_2. We may assume therefore that there is a circuit C_1 of H_1 with vertices the vertices of $V(H_1) \cap \mathrm{bd}(\Delta)$, in order, and similarly for C_2. Let r_1 be as before. Then r_1 is incident with the edge joining u, v because H_1 is rigid; and so, if we extend Δ to a sphere Σ, it follows that $U(C_1)$ is the boundary of the region s_1 of the drawing H_1 in Σ which includes $\Sigma - \Delta$. Add a new vertex v drawn in s_1, adjacent to every vertex of C_1, and let the enlarged graph be J_1; and define J_2 similarly. Then J_1 is a simple, 3-connected drawing in a sphere and J_2 is an isomorphic drawing, and the result follows from Whitney's theorem [7]. □

(6.5) *Let H be a rigid drawing in a closed disc Δ. Let H be a subgraph of G, let $\mathscr{S}$ be a 3-segregation of G with $V(H) \cap bd(\Delta) \subseteq V(\mathscr{S})$, and let α be an arrangement of $\mathscr{S}$ in Δ such that $\alpha(v) = v$ for all $v \in V(H) \cap bd(\Delta)$. Then there is an arrangement α' of $\mathscr{S}$ in Δ such that α' is compatible with H.*

Proof. First we observe the following.

(1) *For every circuit C of H there are three disjoint paths of H from $V(C)$ to $V(H) \cap bd(\Delta)$.*

[p. 35] *Subproof.* Otherwise, by a form of Menger's theorem for planar graphs, there is an O-arc $F \subseteq \Delta$ with $F \cap U(H) \subseteq V(H)$ and $|F \cap V(H)| \leq 2$, bounding a closed disc in Δ including $U(C)$. But H is rigid, and F only bounds one closed disc in Δ, a contradiction.

Let $(A, \Omega) \in \mathscr{S}$, and let J be a graph obtained from $A \cap H$ by adding a new vertex adjacent to all the (at most three) vertices in $V(H) \cap \bar{\Omega}$. We claim that

(2) *J is planar.*

Subproof. We may assume that $A \cap H$ has a circuit C, for otherwise J is clearly planar. By (1), there are three disjoint paths of H from $V(C)$ to $V(H) \cap bd(\Delta)$, and since each of these paths has a unique vertex in $V(H) \cap \bar{\Omega}$, we deduce that there are three disjoint paths of H from $V(H) \cap \bar{\Omega}$ to $V(H) \cap bd(\Delta)$. But the graph obtained from H by adding a new vertex adjacent to every vertex in $V(H) \cap bd(\Delta)$ is planar, and it follows that J is planar, as required.

From (2), we deduce that $A \cap H$ can be drawn in the closed disc $\alpha(A, \Omega)$ in such a way that for each $v \in V(H) \cap \bar{\Omega}$, v is represented in this drawing by $\alpha(v)$, and the only vertices of $A \cap H$ drawn in $bd(\alpha(A, \Omega))$[F6] are those in $V(H) \cap \bar{\Omega}$. By taking the union of these drawings over all (A, Ω), after first moving the discs $\alpha(A, \Omega)$ away from $bd(\Delta)$ at every vertex of $V(\mathscr{S})$ not in $bd(\Delta)$,[F7] we deduce that there is a drawing H' in Δ with $V(H') \cap bd(\Delta) = V(H) \cap bd(\Delta)$, and an isomorphism $\phi : H \to H'$, such that

(i) $\alpha(v) = \phi(v)$ for all $v \in V(H) \cap V(\mathscr{S})$, and

(ii) for each $(A, \Omega) \in \mathscr{S}$, $U(H') \cap bd(\alpha(A, \Omega)) = \phi(V(H) \cap \bar{\Omega})$, and $H' \cap \alpha(A, \Omega) = \phi(H \cap A)$.

From (i) it follows that $\phi(v) = v$ for all $v \in V(H) \cap bd(\Delta)$, since $\alpha(v) = v$ for all $v \in V(H) \cap bd(\Delta)$. Since H is rigid, there is a homeomorphism $\psi : \Delta \to \Delta$ extending ϕ, by (6.4). For each $(A, \Omega) \in \mathscr{S}$, let $\alpha'(A, \Omega) = \psi^{-1}(\alpha(A, \Omega))$, and for each $v \in V(\mathscr{S})$, let $\alpha'(v) = \psi^{-1}(\alpha(v))$; then the theorem is satisfied. □

7. Distance and Freedom

We shall need the following, which is [5, theorem (8.12)].

(7.1) *Let H be a 2-cell drawing with $|E(H)| \geq 2$ in a surface Σ, and let $\mathscr{T}$ be a respectful tangle in H of order $\theta \geq 1$, with metric d. Then for every $z \in A(H)$ there exists $e \in E(H)$ with $d(z, e) = \theta$.*

Also, from [5, theorem (6.3)] it follows that

[p. 36] (7.2) *Let H be a 2-cell drawing in Σ with $E(H) \neq \emptyset$, and let $\mathscr{T}$ be a respectful tangle in H. Let K be a radial drawing of H, and let $(A, B) \in \mathscr{T}$. Then for every edge e of A there is a circuit D of K with $V(D) \cap V(H) \subseteq V(A \cap B)$ and with $e \subseteq ins(D)$.*

[F6] **Misprint.** A misprint for $bd(\alpha(A, \Omega))$.

[F7] **Proof gap.** A gap in the proof: the definition of an arrangement (page 30) allows $\alpha(v) \in bd(\Delta)$ for a vertex v of $V(H) \cap V(\mathscr{S})$ not in $bd(\Delta)$. Then (i) puts $\phi(v)$ in $bd(\Delta)$, against $V(H') \cap bd(\Delta) = V(H) \cap bd(\Delta)$. Moving the discs inward first removes this case. With this step, (6.5) holds as printed.

We deduce

(7.3) *Let H be a 2-cell drawing in Σ with $E(H) \neq \emptyset$, and let $\mathscr{T}$ be a respectful tangle in H of order θ, with metric d. Let $z \in A(H)$, and let $(A, B) \in \mathscr{T}$ with $E(B) \neq \{z\}$. Then there exists $e \in E(B)$ with $d(z, e) \geq \theta - |V(A \cap B)|$.*

Proof. We proceed by induction on $|V(B)|$. If there exists $v \in V(A \cap B)$ such that $(A, B \setminus v)$ is a separation of G, then $(A, B \setminus v) \in \mathscr{T}$ and the result follows from the inductive hypothesis. We may assume then that for every $v \in V(A \cap B)$ there is an edge of B incident with v.

If $|E(H)| = 1$ then $\theta = 1$ and $B = H$, and $z \notin E(H)$ (since $E(B) \neq \{z\}$), and so $d(z, e) \geq 1 \geq \theta - |V(A \cap B)|$ where $E(H) = \{e\}$, as required. We assume then that $|E(H)| \geq 2$. By (7.1) there exists $e \in E(H)$ with $d(z, e) = \theta$. If $e \in E(B)$ we are finished, and so we may assume that $e \in E(A)$. Let K be a radial graph of H. By (7.2) there is a circuit D of K with $V(D) \cap V(H) \subseteq V(A \cap B)$ and which $e \subseteq ins(D)$. Choose $v \in V(D) \cap V(H)$. Then

$$d(v, e) \leq |V(D) \cap V(H)| \leq |V(A \cap B)|$$

and so $d(z, v) \geq \theta - |V(A \cap B)|$. Since $v \in V(D) \cap V(H) \subseteq V(A \cap B)$, there is an edge f of B incident with v. If $z \neq f$, then $d(z, f) \geq d(z, v) \geq \theta - |V(A \cap B)|$, as required. If $z = f$, since $E(B) \neq \{z\}$ there exists $g \in E(B) - \{f\}$; and then $d(z, g) \geq \cancel{1 \geq} d(z, v) \geq$ [F8] $\theta - |V(A \cap B)|$ as required. □

(7.4) *Let H be a 2-cell drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H. Let K be a radial drawing of H, let $(A, B) \in \mathscr{T}$, and let C be a circuit of A with $|V(B \cap C)| \leq 1$. Then there is a circuit D of K with $V(D) \cap V(H) \subseteq V(A \cap B)$ and $U(C) \subseteq ins(D)$.*

Proof. Let $e \in E(C)$. Then $e \in E(A)$, and by (7.2) there is a circuit D of K with $V(D) \cap V(H) \subseteq V(A \cap B)$ and with $e \subseteq ins(D)$. Since

$$U(C) \cap U(D) = U(C) \cap (V(D) \cap V(H)) \subseteq U(C) \cap V(B)$$

it follows that $|U(C) \cap U(D)| \leq 1$, and so $U(C) \subseteq ins(D)$, as required. □

[F8] **Proof gap.** The step $1 \geq d(z, v)$ is false in general: the atom of the edge f is a region of the radial drawing bounded by a circuit of length 4, so $d(f, v)$ can be 2. But $d(z, g) \geq d(z, v)$ still holds. That circuit has v and the atom of z in its inside, so $d(z, v) \leq 2$. A closed walk of the radial drawing whose inside contains the atoms of both z and g cannot lie in the closure of the atom of z, so it has length at least 4, and $d(z, g) \geq 2$. So (7.3) holds as printed.

(7.5) *Let H be a 2-cell drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H of order θ. Let $\lambda \geq 0$ be an integer, and let $\Lambda \subseteq \Sigma$ be an λ-zone. Let K be* [p. 37] *a radial drawing of H, and let D be a circuit of K with $U(D) \subseteq \bar{\Lambda}$ and $|E(D)| < 2(\theta - \lambda)$. Then $ins(D) \subseteq \bar{\Lambda}$.*

Proof. Choose $z \in A(H)$ so that Λ is a λ-zone around z. Since H and K both have circuits it follows that $|E(H)| \geq 2$. By (7.1) there exists $e \in E(H)$ with $d(z, e) = \theta$. Let $\Delta \subseteq \bar{\Lambda}$ be the closed disc bounded by $U(D)$; then $e \cap \Delta \subseteq e \cap \bar{\Lambda} = \emptyset$, and so Δ is the unique closed disc in Σ which is bounded by $U(D)$ and disjoint from e. Now $ins(D)$ is a closed disc in Σ bounded by $U(D)$, and we claim that $ins(D) \cap e = \emptyset$. For let $v \in V(D) \cap V(H)$. Then $d(z, v) \leq \lambda$ since $v \in U(D) \subseteq \bar{\Lambda}$, and so $d(v, e) \geq \theta - \lambda$ since $d(z, e) = \theta$; and hence $ins(D) \cap e = \emptyset$, since $\frac{1}{2}|E(D)| < \theta - \lambda$ and $v \in ins(D)$. From the uniqueness of Δ we deduce that $ins(D) = \Delta$, and so $ins(D) \subseteq \bar{\Lambda}$, as required. □

(7.6) *Let H be a 2-cell drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H of order θ. Let $\lambda \geq 0$ be an integer, and let $\Lambda \subseteq \Sigma$ be a λ-zone. Let K be a radial drawing of H, and let D be a circuit of K with $|E(D)| < 2(\theta - \lambda)$ and $bd(\Lambda) \subseteq ins(D)$. Then $\bar{\Lambda} \subseteq ins(D)$.*

Proof. Let $z \in A(K)$ be such that Λ is a λ-zone around z. Let Λ' be the open disc bounded by $bd(\Lambda)$ in $ins(D)$, and suppose that $\Lambda' \neq \Lambda$. Then $bd(\Lambda)$ bounds two open discs, and so Σ is a sphere, and $\Sigma = \bar{\Lambda} \cup \Lambda'$. Moreover, $U(D) \cap \Lambda' = \emptyset$ because $\Lambda' \subseteq ins(D)$, and so $U(D) \subseteq \bar{\Lambda}$. By (7.5), $ins(D) \subseteq \bar{\Lambda}$, and so $\Lambda' \subseteq \bar{\Lambda}$, a contradiction. Thus $\Lambda' = \Lambda$, and so $\bar{\Lambda} \subseteq ins(D)$, as required. □

If α is an arrangement in a surface Σ of a segregation $\mathscr{S}$ of G, a subset $X \subseteq \Sigma$ is said to be α-normal if

$$X \cap \alpha(A, \Omega) \subseteq \{\alpha(v) : v \in \bar{\Omega}\}$$

for all $(A, \Omega) \in \mathscr{S}$.

(7.7) *Let H be a 2-cell drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H of order $\theta \geq 6$. Let $z \in A(H)$, let C_1, C_2 be disjoint circuits of H, and let $U(C_1)$, $U(C_2)$ bound closed discs Δ_1, $\Delta_2 \subseteq \Sigma$ with the following properties:*

(i) $z \subseteq \Delta_2 - bd(\Delta_2)$, and $\Delta_2 \subseteq \Delta_1 - bd(\Delta_1)$

(ii) $d(z, x) \leq \theta - 4$ for all $x \in A(H)$ with $x \subseteq \Delta_1$

(iii) $x \subseteq \Delta_2 - bd(\Delta_2)$ for all $x \in A(H)$ with $d(z, x) \leq 3$.

Let H be a subgraph of G, let $\mathscr{S}$ be a $\mathscr{T}$-local 3-segregation of G, and let α be an arrangement of $\mathscr{S}$ in Σ compatible with H. Then there is an α-normal O-arc $F \subseteq \Delta_1$, bounding a closed disc in Δ_1 including Δ_2.

[p. 38] *Proof.* For each $(A,\Omega)\in\mathscr{S}$, let

$$J(A,\Omega)=\begin{cases}(bd(\alpha(A,\Omega)),\{\alpha(v):v\in\bar{\Omega}\}) \text{ if } \bar{\Omega}\neq\emptyset\\ (bd(\alpha(A,\Omega)),\{x\}) \text{ for some } x\in bd(\alpha(A,\Omega)), \text{ if } \bar{\Omega}=\emptyset.\end{cases}$$

Then each $J(A,\Omega)$ is a drawing in Σ, and so is $J=\bigcup(J(A,\Omega):(A,\Omega)\in\mathscr{S})$. We claim

(1) *$r\subseteq\Delta_1-bd(\Delta_1)$ for every region r of J with $\bar{r}\cap(\Delta_2-bd(\Delta_2))\neq\emptyset$.*

Subproof. Suppose that for some region r of J, $\bar{r}\cap(\Delta_2-bd(\Delta_2))\neq\emptyset$ and $r\not\subseteq\Delta_1-bd(\Delta_1)$. Since $\Delta_2\subseteq\Delta_1-bd(\Delta_1)$ it follows that $r\cap e_1\neq\emptyset$ for some $e_1\in E(C_1)$; and since $U(J)\cap e_1=\emptyset$, it follows that $e_1\subseteq r$. Similarly, since $\bar{r}\cap(\Delta_2-bd(\Delta_2))\neq\emptyset$, there exists $e_2\in E(C_2)$ with $e_2\subseteq r$. Choose $(A,\Omega)\in\mathscr{S}$ with $e_1\in E(A)$. Since $\alpha(A,\Omega)-bd(\alpha(A,\Omega))$ is a region of J including e_1 (because α is compatible with H) it follows that $r=\alpha(A,\Omega)-bd(\alpha(A,\Omega))$. Since $e_2\subseteq r$ it follows that $e_2\in E(A)$ (because α is compatible with H). Since C_1, C_2 are disjoint and both have an edge in A and $|\bar{\Omega}|\leq 3$, it follows that for some $i(i=1$ or $2)$, $C_i\subseteq A$ and $|V(C_i)\cap\bar{\Omega}|\leq 1$. Let (A,B) be the separation of G with $V(A\cap B)=\bar{\Omega}$; then $(A\cap H,B\cap H)\in\mathscr{T}$ since $\mathscr{S}$ is $\mathscr{T}$-local. Since $|V(C_i\cap B\cap H)|\leq 1$ and C_i is a circuit of $A\cap H$, it follows from (7.4) that there is a circuit D of a radial drawing K of H with $V(D)\cap V(H)\subseteq V(A\cap B\cap H)$ and $U(C_i)\subseteq ins(D)$. Since $U(C_i)\subseteq ins(D)$ and Δ_i is a $(\theta-4)$-zone, it follows from (7.6) (taking $\lambda=\theta-4$) that $\Delta_i\subseteq ins(D)$, and so $z\subseteq ins(D)$. But this contradicts (iii), since $ins(D)\not\subseteq\Delta_2-bd(\Delta_2)$. This proves (1).

By (5.2) applied to J (taking $Z=\Delta_2-bd(\Delta_2)$) and (1), there is a circuit C of J with $U(C)\subseteq\Delta_1$, bounding an open disc in Δ_1 including $\Delta_2-bd(\Delta_2)$. By perturbing $U(C)$ slightly so that it becomes α-normal, we obtain an O-arc satisfying the theorem. □

(7.8) *Let $\beta\geq 4$ be even, let $\lambda\geq 0$, and let $\theta\geq 4\lambda+2$. Let H be a 2-cell drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H of order θ. Let $z\in A(H)$, let C_1, C_2 be disjoint circuits of H, and let $U(C_1)$, $U(C_2)$ bound closed discs Δ_1, $\Delta_2\subseteq\Sigma$ with the following properties:*

(i) $z\subseteq\Delta_2-bd(\Delta_2)$, and $\Delta_2\subseteq\Delta_1-bd(\Delta_1)$

(ii) $\Delta_1-bd(\Delta_1)$ is a $(\theta-\beta-4\lambda-2)$-zone around z

(iii) $\Delta_2-bd(\Delta_2)$ is a λ-zone around z

(iv) $x\subseteq\Delta_1-bd(\Delta_1)$ for every $x\in A(H)$ with $d(z,x)<\beta$, and

(v) *there are circuits $C_3,\ldots,C_{\beta/2}$ of $H\cap\Delta_1$ such that C_1, C_2, …, $C_{\beta/2}$ are mutually disjoint and $U(C_3),\ldots,U(C_{\beta/2})$ all bound closed discs in Δ_1 including Δ_2.*

[p. 39] *Let $P_1, \dots, P_\beta$ be disjoint paths of H from $V(C_2)$ to $V(C_1)$, such that for $1 \le i \le \beta$, P_i has initial vertex $v_i \in V(C_2)$ and $U(P_i) \cap (\Delta_2 - bd(\Delta_2)) = \emptyset$. Then $\{v_1, \dots, v_\beta\}$ is free with respect to the tangle of order $\theta - 4\lambda - 2$ obtained from $\mathscr{T}$ by clearing the λ-zone $\Delta_2 - bd(\Delta_2)$.*

Proof. Let $H' = H \cap (\Sigma - (\Delta_2 - bd(\Delta_2)))$, and let $\mathscr{T}'$ be the tangle of order $\theta - 4\lambda - 2$ in H' obtained by clearing $\Delta_2 - bd(\Delta_2)$. Suppose that $\{v_1, \dots, v_\beta\}$ is not free with respect to $\mathscr{T}'$, and choose $(A, B) \in \mathscr{T}'$ of order $< \beta$ with $v_1, \dots, v_\beta \subseteq V(A)$. This is possible, for certainly $\beta \le \theta - 4\lambda - 2$. Since $|V(A \cap B)| < \beta$ and $P_1, \dots, P_\beta$ are mutually disjoint, there exists j with $1 \le j \le \beta$ such that $V(P_j) \cap V(A \cap B) = \emptyset$. Since P_j has an end in $V(A)$ and P_j is a connected subgraph of H', it follows that $V(P_j) \subseteq V(A) - V(B)$.

Since $|V(A \cap B)| < \beta$ there exists i with $1 \le i \le \beta/2$ such that $|V(C_i \cap A \cap B)| \le 1$, because $C_1, \dots, C_{\beta/2}$ are mutually disjoint. Since $U(C_i)$ bounds a disc Δ in Δ_1 including Δ_2, and P_j is a path from $V(C_2)$ to $V(C_1)$, it follows that $V(C_i \cap P_j) \ne \emptyset$. Since $V(P_j) \subseteq V(A) - V(B)$ we deduce that $V(C_i) \not\subseteq V(B)$. Since $|V(C_i \cap A \cap B)| \le 1$ it follows that $C_i \subseteq A$ and $|V(C_i \cap B)| \le 1$. By (7.4) there is a circuit D of a radial drawing K' of H' such that $V(D) \cap V(H') \subseteq V(A \cap B)$ and $U(C_i) \subseteq ins'(D)$, where ins' is the inside function of $\mathscr{T}'$. Now $\Delta - bd(\Delta)$ is a $(\theta - \beta - 4\lambda - 2)$-zone around z (in $H, \mathscr{T}$), and hence is a $(\theta - \beta - 4\lambda - 2)$-zone around the region $\Delta_2 - bd(\Delta_2)$ (in $H', \mathscr{T}'$). Since $|E(D)| < 2\beta$ and $bd(\Delta) \subseteq ins'(D)$, it follows from (7.6) applied to $H', \mathscr{T}'$ that $\Delta \subseteq ins'(D)$. In particular, $U(D) \cap \Delta_2 \subseteq bd(\Delta_2)$. Choose a radial drawing K of H so that $K' \setminus v$ is a subgraph of K, where v is the unique vertex of K' in $\Delta_2 - bd(\Delta_2)$; then $D \subseteq K$. Since $|U(D) \cap V(H)| < \beta \le \theta - 4\lambda - 2$, it follows that $ins'(D) = ins(D)$. Since $z \in \Delta_2 \subseteq \Delta \subseteq ins'(D) = ins(D)$, we deduce from (iv) that $ins(D) \subseteq \Delta_1 - bd(\Delta_1)$. But there is a vertex of P_j in $ins(D)$, and $U(P_j) \cap U(D) = \emptyset$ because $V(P_j) \cap V(A \cap B) = \emptyset$, and so $U(P_j) \subseteq ins(D) \subseteq \Delta_1 - bd(\Delta_1)$. Yet P_j has an end in C_1, a contradiction.

Hence $\{v_1, \dots, v_\beta\}$ is free with respect to $\mathscr{T}'$, as required. □

8. Segregation In A Rural Neighbourhood

Now we develop some lemmas about societies. If Ω is a cyclic permutation, let us write $\bar{\Omega} = \{v_1, \dots, v_n\}$, numbered in order under Ω. An Ω-interval is a set $\{v_j : i \le j \le k\}$ where $0 \le i \le k \le n$, or the complement in $\bar{\Omega}$ of such a set. (This definition is independent of which element of $\bar{\Omega}$ is chosen to be v_1). A *transaction* in a society (G, Ω) is a set $\mathscr{P}$ of mutually disjoint paths of G, each with distinct ends both in $\bar{\Omega}$, such that there is an Ω-interval containing exactly one end of each member of $\mathscr{P}$. If $\mathscr{P}$ is a trans-

action, $\bigcup \mathscr{P}$ denotes the union of the members of $\mathscr{P}$, and $V(\mathscr{P})$ denotes the set of ends [p. 40] of members of $\mathscr{P}$. Thus, $V(\mathscr{P}) \subseteq \bar{\Omega}$. We say that $\mathscr{P}$ is *crooked* if for every $P \in \mathscr{P}$ with ends u, v say, there exist $u', v' \in V(\mathscr{P})$ such that u, u', v, v' are distinct and occur in that order in Ω. A *cross* is a crooked transaction of cardinality 2.

Let α be an arrangement in a surface Σ of segregation $\mathscr{S}$ of G, and let $F \subseteq \Sigma$ be an α-normal O-arc. Let $X = \{v \in V(\mathscr{S}) : \alpha(v) \in F\}$ and $X' = \{\alpha(v) : v \in X\}$. Let Ω' be a natural order of X' from F, and let Ω be the cyclic permutation with $\bar{\Omega} = X$ mapped to Ω' by α. We write $\Omega = \alpha^{-1}(F)$, for convenience. The proof of [3, theorem (11.11)] may easily be adapted to yield the following.

(8.1) *A society (G, Ω) has no cross if and only if there is a 3-segregation $\mathscr{S}$ of G with $\bar{\Omega} \subseteq V(\mathscr{S})$ and an arrangement α of $\mathscr{S}$ in a closed disc Δ with $\Omega = \alpha^{-1}(bd(\Delta))$.*

A *neighbourhood* is a triple (G, Ω_1, Ω_2) where G is a graph and Ω_1, Ω_2 are cyclic permutations in $V(G)$. If (G, Ω_1) is a society and (G_1, G_2) is a separation of G with $\bar{\Omega}_1 \subseteq V(G_1)$, and Ω_2 is a cyclic permutation with $\bar{\Omega}_2 = V(G_1 \cap G_2)$, we say that (G, Ω_1) is the *composition* of (G_2, Ω_2) and $(G_1, \Omega_1, \Omega_2)$. A neighbourhood (H, Ω_1, Ω_2) is *rural* if H is a drawing in a closed cylinder Σ with ends F_1, F_2 and Ω_i is a natural order of $V(H) \cap F_i$ from F_i for $i = 1, 2$. (The *ends* of a closed cylinder Σ are the two O-arcs in $bd(\Sigma)$.) This definition of "rural" is slightly different from that in [3], where Σ was not required to be a manifold.

Let (H, Ω_1, Ω_2) be a rural neighbourhood, in a closed cylinder Σ with ends F_1, F_2. It has *breadth* $\geq q$ if there are q circuits of H, mutually disjoint, and each winding around Σ, in the natural sense. The following is [3, theorem (10.4)].

(8.2) *Let $\pi \geq 0$ be an integer with $\pi \neq 1, 3$, and let (G, Ω_1) be the composition of a society (G_2, Ω_2) with a rural neighbourhood $(G_1, \Omega_1, \Omega_2)$ of breadth $\geq \pi + 5$. Suppose that $X \subseteq \bar{\Omega}_1$ and there are $2\pi + 8$ mutually disjoint paths of G_1, each from X to $\bar{\Omega}_2$ with no vertex in $\bar{\Omega}_1$ except its initial vertex. If there is a crooked transaction $\mathscr{P}$ in (G, Ω_1) with cardinality $\geq \pi$, then $\mathscr{P}$ may be chosen so that in addition $V(\mathscr{P}) \subseteq X$ and no member of $\mathscr{P}$ has any internal vertex in $\bar{\Omega}_1$.*

We remark that, in the case $\pi = 2$ (which, for (4.1), is all we need of (8.2)), one can improve (8.2) numerically as follows.

(8.3) *Let (G, Ω_1) be the composition of a society (G_2, Ω_2) with a rural neighbourhood $(G_1, \Omega_1, \Omega_2)$ of breadth ≥ 3. Suppose that $X \subseteq \bar{\Omega}_1$ and there are 4 mutually disjoint paths in G_1, each from X to $\bar{\Omega}_2$ with no vertex in*

$\bar{\Omega}_1$ *except its initial vertex. If there is a cross* $\mathscr{P}$ *in* (G, Ω_1) *then* $\mathscr{P}$ *may be* [p. 41] *chosen so that in addition* $V(\mathscr{P}) \subseteq X$ *and no member of* $\mathscr{P}$ *has any internal vertex in* $\bar{\Omega}$.

However, ~~we omit the proof, and~~ we shall not use the result here.

We give the proof. In it we use the terms of [3], as in (8.2), and also the notation of [3], which is not related to the notation of (8.3). Thus if $(G \cup G_0, \Omega)$ is the composition of a society (G_0, Ω_0) with a neighbourhood (G, Ω, Ω_0), then G and G_0 have no common edge and $V(G \cap G_0) = \bar{\Omega}_0$. We shall prove the following.

(1) *Let* (G, Ω, Ω_0) *be a rural neighbourhood of breadth* ≥ 3, *let* $X \subseteq \bar{\Omega}$ *be linked with* $|X| \geq 4$, *and let* $(G \cup G_0, \Omega)$ *be the composition of a society* (G_0, Ω_0) *with* (G, Ω, Ω_0). *If* $(G \cup G_0, \Omega)$ *has a cross, then it has a cross* $\mathscr{P}$ *with* $V(\mathscr{P}) \subseteq X$ *such that no member of* $\mathscr{P}$ *has an internal vertex in* $\bar{\Omega}$.

This gives (8.3), applied to the neighbourhood $(G_1, \Omega_1, \Omega_2)$, the society (G_2, Ω_2), and the set X' of initial vertices of the four given paths. Indeed $X' \subseteq X$ and $|X'| = 4$, since the paths are disjoint, and X' is linked, by the four paths. Now (1) is the case $p = 2$ of [3, (10.4)], with breadth ≥ 3 in place of $\geq p + 5$ and $|X| \geq 4$ in place of $|X| \geq 2p + 8$. The proof of [3, (10.4)] uses [3, (10.3)], whose proof uses [3, (5.2)], and only [3, (5.2)] needs $p + 4$ separating paths. We show first that for crosses two separating paths are enough, and then we follow the proofs of [3, (10.3)] and [3, (10.4)].

We use the following fact several times. Let (H_1, H_2) be a separation of a graph, and let R be a connected subgraph with a vertex in $V(H_1)$ and no vertex in $V(H_1 \cap H_2)$. Then $R \subseteq H_1$. For an edge of R with an end in $V(H_1) - V(H_2)$ is not an edge of H_2, and so it is an edge of H_1, and its other end is in $V(H_1)$ and hence in $V(H_1) - V(H_2)$; since R is connected, every edge and every vertex of R is in H_1.

(2) *Let* (G, Ω) *be a society with a cross. Let* $\{Q_1, Q_2\}$ *be a transaction in* (G, Ω) *such that* Q_1, Q_2 *are separating and* Q_i *has ends* a_i, b_i $(i = 1, 2)$, *where* (a_2, a_1, b_1, b_2) *is clockwise. Let* (H_1, H_2) *be a separation of* G *with* $V(H_1 \cap H_2) = V(Q_2)$, $Q_2 \subseteq H_1$, $\Omega(a_2, b_2) \subseteq V(H_1)$ *and* $\Omega(b_2, a_2) \subseteq V(H_2)$. *Then either there is a cross* $\{P, P'\}$ *of* (G, Ω) *with* $P, P' \subseteq H_1$, *or there is a cross* $\mathscr{P}$ *of* (G, Ω) *with* $\Omega(a_1, b_1) \cap V(\mathscr{P}) = \emptyset$.

Since (a_2, a_1, b_1, b_2) is clockwise, the vertices a_2 and b_2 are in $\Omega(b_1, a_1)$, and $\Omega(a_1, b_1) \subseteq \Omega(a_2, b_2)$. Suppose first that Q_1, Q_2 have no internal vertex in $\bar{\Omega}$. A cross is a crooked transaction of cardinality 2, and so we may choose $\mathscr{P}$ and a fence for it such that the statement **S** of [3, Section 4] holds with $\mathscr{Q} = \{Q_1, Q_2\}$ and $p = 2$. Then $|\mathscr{P}| = 2$. For otherwise, by [3, (3.4)], some subset of $\mathscr{P}$ with two members is a crooked transaction; its union with $\bigcup \mathscr{Q}$ is included in $\bigcup \mathscr{P} \cup \bigcup \mathscr{Q}$, and so it is equal to it by (i) of **S**, and then (ii) of

S is false. Hence $\mathscr{P}$ is a cross, and $|V(\mathscr{P})| = 4$. If $V(\mathscr{P}) \cap \Omega(a_1, b_1) = \emptyset$ the second outcome of (2) holds, and so we may assume that there is a vertex $x_0 \in V(\mathscr{P}) \cap \Omega(a_1, b_1)$.

(3) *For $j = 1, 2$, if P is an Ω-path with both ends in $\Omega(a_j, b_j)$ and no vertex in $V(Q_j)$, then $P \subseteq H_1$.*

The ends of P are in $\Omega(a_2, b_2) \subseteq V(H_1)$. We claim that P has no vertex in $V(Q_2)$. For $j = 2$ this is a hypothesis. Let $j = 1$, and suppose that P meets $V(Q_2)$. Let R_1 be a minimal subpath of P between an end u of P and a vertex $w \in V(Q_2)$, and let R_2 be the subpath of Q_2 between w and a_2. Then $R_1 \cup R_2$ is a path between u and a_2, since R_1 meets $V(Q_2)$ only in w. It has no vertex in $V(Q_1)$, since P has none and Q_1, Q_2 are disjoint. But $u \in \Omega(a_1, b_1)$ and $a_2 \in \Omega(b_1, a_1)$, and so $R_1 \cup R_2$ is an Ω-path that crosses Q_1, which is impossible since Q_1 is separating. This proves our claim. Since $V(H_1 \cap H_2) = V(Q_2)$, the fact above shows that $P \subseteq H_1$. This proves (3).

Suppose that $Q_j \cap \bigcup \mathscr{P}$ is null for some $j \in \{1, 2\}$. Since Q_j is separating and $x_0 \in \Omega(a_1, b_1) \subseteq \Omega(a_j, b_j)$, [3, (5.1)] shows that $\mathscr{P}$ has a subset which is a cross with all four ends in $\Omega(a_j, b_j)$. This subset is $\mathscr{P}$ itself, and by (3) both members of $\mathscr{P}$ are subgraphs of H_1, so the first outcome of (2) holds. We may therefore assume that Q_1 and Q_2 both meet $\bigcup \mathscr{P}$. Moreover $\mathscr{P}$ has no overpass, since removing a member of $\mathscr{P}$ leaves a transaction with one member, which does not have two peripheral members. So by [3, (4.3)], each of Q_1, Q_2 either has both ends in $V(\mathscr{P})$ or is half-covered.

(4) *If Q_j is half-covered, where $j \in \{1, 2\}$, then $\mathscr{P} = \{P, P'\}$, where P and Q_j have a common end s, $P \cap Q_j$ is a path between s and some vertex c, P' has no vertex in $V(Q_j)$, and the three vertices in $V(\mathscr{P}) - \{s\}$ are either all in $\Omega(a_j, b_j)$ or all in $\Omega(b_j, a_j)$.*

For let P be the member of $\mathscr{P}$ that meets Q_j, as in [3, (4.3)(iii)], let s be the common end of P, Q_j and $P \cap Q_j$, and let P' be the other member of $\mathscr{P}$. Then P' has no vertex in $V(Q_j)$, since Q_j meets only one member of $\mathscr{P}$. Since Q_j is separating, P' does not cross Q_j, and so both ends of P' are in $\Omega(a_j, b_j)$ or both are in $\Omega(b_j, a_j)$. We assume the first; the second case is the same with the two sides exchanged. Let the ends of P' be u, w, where (a_j, u, w, b_j) is clockwise, and let t be the end of P other than s. Since $\mathscr{P}$ is crooked, P' is not peripheral, and so $\Omega(u, w)$ contains s or t. It does not contain s, since $s \in \{a_j, b_j\}$. Hence $t \in \Omega(u, w) \subseteq \Omega(a_j, b_j)$, and so u, w, t are all in $\Omega(a_j, b_j)$. This proves (4).

Since Q_2 meets $\bigcup \mathscr{P}$, it has an end in $V(\mathscr{P})$, by (4) if Q_2 is half-covered. This end is in $\Omega(b_1, a_1)$, while $x_0 \in \Omega(a_1, b_1)$. So for $s \in \{a_1, b_1\}$ the set $V(\mathscr{P}) - \{s\}$ meets both $\Omega(a_1, b_1)$ and $\Omega(b_1, a_1)$, and hence Q_1 is not half-covered, by (4). Thus $a_1, b_1 \in V(\mathscr{P})$. If also $a_2, b_2 \in V(\mathscr{P})$, then

x_0, a_1, b_1, a_2, b_2 are five distinct members of $V(\mathscr{P})$, which is impossible. Hence Q_2 is half-covered. Let P, P', s, c be as in (4) with $j = 2$, and let t be the end of P other than s. Since $x_0 \in \Omega(a_2, b_2)$ and $x_0 \neq s$, the three vertices in $V(\mathscr{P}) - \{s\}$ are in $\Omega(a_2, b_2)$, by (4). In particular $P' \subseteq H_1$, by (3). Since $P \cap Q_2$ is the subpath of P between s and c, the subpath R of P between c and t meets $V(Q_2)$ only in c. Now $t \neq c$, since $t \in \Omega(a_2, b_2)$, while the only vertices of Q_2 in $\bar{\Omega}$ are a_2 and b_2. So $R \setminus c$ is connected, has no vertex in $V(Q_2) = V(H_1 \cap H_2)$, and contains $t \in V(H_1)$, and by the fact above $R \setminus c \subseteq H_1$. The edge of R at c is then in H_1, since its other end is in $V(H_1) - V(H_2)$, and $P \cap Q_2 \subseteq Q_2 \subseteq H_1$. Hence $P \subseteq H_1$, and the first outcome of (2) holds.

Now we drop the assumption that Q_1, Q_2 have no internal vertex in $\bar{\Omega}$. Let G' be obtained from G by adding, for each $v \in \bar{\Omega}$, a new vertex v' and a new edge vv', and let Ω' be the cyclic permutation with $\bar{\Omega}' = \{v' : v \in \bar{\Omega}\}$ that maps v' to w' whenever Ω maps v to w. For an Ω-path P with ends u, w, let P' be P together with the edges uu' and ww'. Then P' is an Ω'-path with no internal vertex in $\bar{\Omega}'$, and every Ω'-path is P' for exactly one Ω-path P, since the new vertices have valency one. Two Ω-paths are disjoint if and only if the corresponding Ω'-paths are, and $\{u, w\}$ crosses $\{u_1, w_1\}$ in Ω if and only if $\{u', w'\}$ crosses $\{u'_1, w'_1\}$ in Ω'. Hence $P \mapsto P'$ and its inverse map crosses to crosses, transactions to transactions, and separating paths to separating paths. Let H'_1 be obtained from H_1 by adding the new edges at the vertices in $\Omega(a_2, b_2) \cup \{a_2, b_2\}$, with their ends, and let H'_2 be obtained from H_2 by adding the other new edges, with their ends, and the vertices a'_2, b'_2. Then (H'_1, H'_2) is a separation of G' with $V(H'_1 \cap H'_2) = V(H_1 \cap H_2) \cup \{a'_2, b'_2\} = V(Q'_2)$ and $Q'_2 \subseteq H'_1$, and $\Omega'(a'_2, b'_2) \subseteq V(H'_1)$ and $\Omega'(b'_2, a'_2) \subseteq V(H'_2)$. Since (G', Ω') has a cross, and $\{Q'_1, Q'_2\}$ is a transaction of separating paths with no internal vertex in $\bar{\Omega}'$, the case already proved applies to (G', Ω'), $\{Q'_1, Q'_2\}$ and (H'_1, H'_2). If it gives a cross of (G', Ω') whose members are subgraphs of H'_1, then the corresponding cross of (G, Ω) has members that are subgraphs of H_1, since H'_1 with the new vertices deleted is H_1. If it gives a cross $\mathscr{P}'$ with $\Omega'(a'_1, b'_1) \cap V(\mathscr{P}') = \emptyset$, the corresponding cross $\mathscr{P}$ of (G, Ω) has $\Omega(a_1, b_1) \cap V(\mathscr{P}) = \emptyset$. This proves (2).

Using (2) in place of [3, (5.2)], we follow the proof of [3, (10.3)] with two separating paths.

(5) *Let (G, Ω, Ω_0) be a rural neighbourhood of breadth ≥ 2, let $X \subseteq \bar{\Omega}$ be weakly-linked with $|X| \geq 4$, and let $(G \cup G_0, \Omega)$ be the composition of a society (G_0, Ω_0) with (G, Ω, Ω_0). If $(G \cup G_0, \Omega)$ has a cross, then it has a cross $\mathscr{P}$ with $V(\mathscr{P}) \subseteq X$.*

We show, by induction on $|Y - X|$, that for every Y with $X \subseteq Y \subseteq \bar{\Omega}$,

if $(G \cup G_0, \Omega)$ has a cross with all ends in Y then it has one with all ends in X. The case $Y = \bar{\Omega}$ is (5). If $Y = X$ there is nothing to prove, and so we assume that there exists $x \in Y - X$. By the inductive hypothesis, applied to $Y - \{x\}$, it suffices to find a cross with all ends in $Y - \{x\}$. List the members of X in clockwise order, starting just after x. Since $|X| \geq 4$, we may take b_1, b_2 to be the first two and a_1, a_2 to be the last two, and then (a_2, a_1, x, b_1, b_2) is clockwise. The set $\{a_1, a_2, b_1, b_2\}$ is weakly-linked, since it is a subset of X, and so by [3, (10.2)] with $s = 2$ there is a transaction $\{Q_1, Q_2\}$ in (G, Ω) such that Q_i has ends a_i, b_i and meets every path of G from $\Omega(a_i, b_i)$ to $\Omega(b_i, a_i) \cup \bar{\Omega}_0$ $(i = 1, 2)$.

Let $i \in \{1, 2\}$. Let J be the union of the components of $G \setminus V(Q_i)$ that contain a vertex in $\Omega(a_i, b_i)$. Let K_1 consist of J, Q_i, and the edges of G between $V(J)$ and $V(Q_i)$, and let K_2 consist of the vertices of G not in $V(J)$ and the edges of G not in K_1. An edge of G with an end in $V(J)$ has its other end in $V(J) \cup V(Q_i)$, and so it is in K_1. Hence (K_1, K_2) is a separation of G with $V(K_1 \cap K_2) = V(Q_i)$, $Q_i \subseteq K_1$ and $\Omega(a_i, b_i) \subseteq V(K_1)$. Moreover $\Omega(b_i, a_i) \cup \bar{\Omega}_0 \subseteq V(K_2)$, since a vertex of $\Omega(b_i, a_i) \cup \bar{\Omega}_0$ in $V(J)$ would be joined to $\Omega(a_i, b_i)$ by a path of G with no vertex in $V(Q_i)$. Since $V(K_1) \cap V(G_0) \subseteq \bar{\Omega}_0 \subseteq V(K_2)$, and G_0 has no edge in G, it follows that $(K_1, K_2 \cup G_0)$ is a separation of $G \cup G_0$ with $V(K_1 \cap (K_2 \cup G_0)) = V(Q_i)$. We call it the separation for Q_i.

Let Ω_Y be the cyclic permutation with $\bar{\Omega}_Y = Y$ that orders Y as Ω does. The crosses of $(G \cup G_0, \Omega_Y)$ are the crosses of $(G \cup G_0, \Omega)$ with all ends in Y. Also $\{Q_1, Q_2\}$ is a transaction in $(G \cup G_0, \Omega_Y)$, since its ends are in $X \subseteq Y$, and (a_2, a_1, b_1, b_2) is clockwise in Ω_Y. For $i = 1, 2$ we have $\Omega_Y(a_i, b_i) \subseteq \Omega(a_i, b_i)$ and $\Omega_Y(b_i, a_i) \subseteq \Omega(b_i, a_i)$, and Q_i is separating in $(G \cup G_0, \Omega_Y)$. For suppose that an Ω_Y-path R crosses Q_i, and let $(K_1, K_2 \cup G_0)$ be the separation for Q_i. Then R has no vertex in $V(Q_i)$. It has one end in $\Omega(a_i, b_i) \subseteq V(K_1)$, and so $R \subseteq K_1$ by the fact above. But its other end is in $\Omega(b_i, a_i) - V(Q_i)$, which is disjoint from $V(K_1)$, a contradiction.

Now we apply (2) to $(G \cup G_0, \Omega_Y)$ and $\{Q_1, Q_2\}$, with (H_1, H_2) the separation $(K_1, K_2 \cup G_0)$ for Q_2. Suppose first that (2) gives a cross $\{P, P'\}$ with $P, P' \subseteq K_1$. Then $\{P, P'\}$ is a cross of (G, Ω), since $K_1 \subseteq G$ and its ends are in $Y \subseteq \bar{\Omega}$. But (G, Ω) has no cross, since it is rural, as in the proof of [3, (10.3)], a contradiction. Hence (2) gives a cross $\mathscr{P}$ of $(G \cup G_0, \Omega_Y)$ with no end in $\Omega_Y(a_1, b_1)$. Its ends are in Y, and x is not one of them since $x \in \Omega_Y(a_1, b_1)$. So $\mathscr{P}$ is a cross of $(G \cup G_0, \Omega)$ with all ends in $Y - \{x\}$. This proves (5).

Now we prove (1), following the proof of [3, (10.4)] with a 3-nest in place of the $(p + 5)$-nest and (5) in place of [3, (10.3)]. Take a presentation

$(\Sigma, \Gamma, \Delta, \Delta_0)$ of (G, Ω, Ω_0) with a 3-nest (C'_1, C'_2, C'_3), and let Δ'_j be the closed disc bounded by the drawing of C'_j $(j = 1, 2, 3)$. The drawing of C'_2 is disjoint from that of C'_3, and so Δ'_2 lies in the interior of Δ'_3 and contains no point of $bd(\Delta)$. Hence no vertex of C'_1 or C'_2 is in $\bar{\Omega}$, and $\bar{\Omega}_0$, which is drawn in $\Delta_0 \subseteq \Delta'_1$, is disjoint from $\bar{\Omega}$. Let $n = |X|$. Since X is linked, there are mutually disjoint $X \to \bar{\Omega}_0$ paths $P'_1, \ldots, P'_n$ of G, each with only its initial vertex in $\bar{\Omega}$. Let G' be obtained from G by deleting every edge of G with an end in $\bar{\Omega}$ except the first edges of $P'_1, \ldots, P'_n$. Then C'_1, C'_2 and $P'_1, \ldots, P'_n$ are subgraphs of G', and $(\Sigma, \Gamma', \Delta, \Delta_0)$ is a presentation of (G', Ω, Ω_0) with the 2-nest (C'_1, C'_2), where Γ' is the drawing of G' given by Γ. By [3, (10.1)] applied to (G', Ω, Ω_0), there is a 2-nest (C_1, C_2) and there are disjoint $X \to \bar{\Omega}_0$ paths $P_1, \ldots, P_n$ of G', such that $P_i \cap C_j$ is a path for $1 \leq i \leq n$ and $j = 1, 2$. Each vertex $v \in \bar{\Omega}$ has valency at most 1 in G'. For the edges of G' at v are first edges of the paths P'_k, the second vertex of each P'_k is an internal vertex or is in $\bar{\Omega}_0$, and so it is not in $\bar{\Omega}$, and only one of these paths starts at v. Hence no vertex of C_1 or C_2 is in $\bar{\Omega}$, since circuits have no vertex of valency 1. For the same reason no internal vertex of P_i is in $\bar{\Omega}$, and so the only vertex of P_i in $\bar{\Omega}$ is its initial vertex x_i, since its other end is in $\bar{\Omega}_0$.

Let Δ_1 be the closed disc bounded by the drawing of C_2. As before, Δ_0 lies in the interior of Δ_1. No vertex in $\bar{\Omega}$ is drawn in Δ_1. For such a vertex is drawn in $bd(\Delta)$, and it is not drawn in the drawing of C_2, since it is not a vertex of C_2, and so it would be in the interior of Δ_1, which is included in the interior of Δ. The drawing of P_i runs from x_i, which is not in Δ_1, to its end in $\bar{\Omega}_0$, which is in Δ_1, and so it meets the drawing of C_2. Since an edge not in C_2 meets the drawing of C_2 only in its ends, P_i has a vertex in $V(C_2)$; let y_i be the first one, and put $Y = \{y_1, \ldots, y_n\}$. Let G_1 be the subgraph of G drawn in Δ_1, and let H_2 be the subgraph of G drawn in the closure of $\Delta - \Delta_1$, without the edges of C_2. An edge of G not in C_2 is drawn in Δ_1 or in the closure of $\Delta - \Delta_1$, and the vertices drawn in $bd(\Delta_1)$ are those of C_2. So (G_1, H_2) is a separation of G with $V(G_1 \cap H_2) = V(C_2)$. Since $V(G_0) \cap V(G) = \bar{\Omega}_0$, which is drawn in the interior of Δ_1, it follows that $(G_1 \cup G_0, H_2)$ is a separation of $G \cup G_0$ with $V((G_1 \cup G_0) \cap H_2) = V(C_2)$, and $\bar{\Omega} \subseteq V(H_2)$. Let Ω_1 be the cyclic permutation with $\bar{\Omega}_1 = V(C_2)$ given by the clockwise orientation of $bd(\Delta_1)$. Then $(G_1, \Omega_1, \Omega_0)$ and (H_2, Ω, Ω_1) are rural neighbourhoods, the first with the 2-nest (C_1, C_2), and $(G \cup G_0, \Omega)$ is the composition of $(G_1 \cup G_0, \Omega_1)$ with (H_2, Ω, Ω_1). By [3, (9.1)] and [3, (3.4)], $(G_1 \cup G_0, \Omega_1)$ has a cross.

The set Y is weakly-linked in $(G_1, \Omega_1, \Omega_0)$, and $|Y| = n \geq 4$. For let $1 \leq i \leq n$. The path $P_i \cap C_2$ contains y_i, and it is a subpath of P_i, and so

y_i is one of its ends, since y_i is the first vertex of P_i in $V(C_2)$. Let z_i be its other end, and let T_i be the subpath of P_i from z_i to the end t_i of P_i in $\bar{\Omega}_0$. Then T_i meets $V(C_2)$ only in z_i, and $t_i \neq z_i$, since t_i is drawn in the interior of Δ_1. So $T_i \setminus z_i$ is a connected subgraph of G with no vertex in $V(C_2)$ and with $t_i \in V(G_1)$, and it is a subgraph of G_1 by the fact above, applied to (G_1, H_2). The edge of T_i at z_i is in G_1 as well, since its other end is in $V(G_1) - V(C_2)$. Hence the subpath of P_i from y_i to t_i, which is the union of $P_i \cap C_2$ and T_i, is a path of G_1. These paths are disjoint, since $P_1, \ldots, P_n$ are, and so Y is weakly-linked.

By (5), applied to $(G_1, \Omega_1, \Omega_0)$, Y and $(G_1 \cup G_0, \Omega_1)$, there is a cross $\{P, P'\}$ of $(G_1 \cup G_0, \Omega_1)$ with all ends in Y. Let y_{i_1}, y_{i_2} be the ends of P and y_{i_3}, y_{i_4} those of P', and for $m = 1, \ldots, 4$ let R_m be the subpath of P_{i_m} between x_{i_m} and y_{i_m}. Then R_m meets $V(C_2)$ only in y_{i_m}, by the choice of y_{i_m}, and so its drawing meets the drawing of C_2 only in y_{i_m}. Since x_{i_m} is not drawn in Δ_1, the rest of R_m is drawn outside Δ_1. So R_m is a path of H_2 from $\bar{\Omega}$ to $\bar{\Omega}_1$, and it meets $G_1 \cup G_0$ only in y_{i_m}, since $V(G_0) \cap V(G) = \bar{\Omega}_0 \subseteq V(G_1)$. Let $P^+ = P \cup R_1 \cup R_2$ and $P'^+ = P' \cup R_3 \cup R_4$. These are disjoint paths of $G \cup G_0$, with ends x_{i_1}, x_{i_2} and x_{i_3}, x_{i_4} in X, since P, P' are disjoint and $R_1, \ldots, R_4$ are disjoint. No internal vertex of P^+ or P'^+ is in $\bar{\Omega}$, since no vertex of $G_1 \cup G_0$ is in $\bar{\Omega}$ and x_{i_m} is the only vertex of R_m in $\bar{\Omega}$. The paths $R_1, \ldots, R_4$ are mutually disjoint, and they are drawn in the closed annulus which is the closure of $\Delta - \Delta_1$, each from one of its boundary circles to the other. So the ends $x_{i_1}, \ldots, x_{i_4}$ occur around $bd(\Delta)$ in the same cyclic order as the ends $y_{i_1}, \ldots, y_{i_4}$ occur around $bd(\Delta_1)$. Since $\{y_{i_1}, y_{i_2}\}$ crosses $\{y_{i_3}, y_{i_4}\}$ in Ω_1, it follows that $\{x_{i_1}, x_{i_2}\}$ crosses $\{x_{i_3}, x_{i_4}\}$ in Ω. Thus $\{P^+, P'^+\}$ is a cross as required in (1). This proves (1), and hence (8.3).[F9]

The main result of this section is the following.

(8.4) *Let Σ be a closed cylinder with ends F_1, F_2, and let H be a rigid drawing in Σ. Let C_1, C_2 be disjoint circuits of H, both winding around Σ, and such that F_2 and $U(C_2)$ both lie in the same component of $\Sigma - U(C_1)$. Let $\pi \geq 0$ with $\pi \neq 1, 3$ and let the regional distance between $U(C_1)$ and $U(C_2)$ be at least $\pi + 4$. Let $X \subseteq V(C_1)$, and let there be $2\pi + 8$ mutually disjoint paths of H from X to $V(C_2)$, each with no vertex in C_1 except its initial vertex. Let H be a subgraph of G_1, let $\mathscr{S}$ be a 3-segregation of G_1, and let α be an arrangement of $\mathscr{S}$ in Σ compatible with H. Let $\Omega_i = \alpha^{-1}(F_i)$ $(i = 1, 2)$, and let Ω_0 be a natural order of $V(C_1)$ from $U(C_1)$. Now let (G, Ω_1) be a society which is the composition of a society (G_2, Ω_2) and the neighbourhood*

[F9] **Missing proof.** The paper omits the proof of (8.3). The proof above is the one used in the formalization.

$(G_1, \Omega_1, \Omega_2)$, and let (G, Ω_1) have a crooked transaction of cardinality $\geq \pi$. Then (G, Ω_0) has a crooked transaction $\mathscr{P}$ of cardinality $\geq \pi$ such that for each $P \in \mathscr{P}$ with ends u, v say, $u, v \in X$ and $U(P \cap H) - \{u, v\}$ is included in the component of $\Sigma - U(C_1)$ which includes $U(C_2)$.

Proof. We proceed by induction on $|V(G_1)| + |E(G_1)|$. We claim that we may assume that

(1) *For each $(A, \Omega) \in \mathscr{S}$, there is no proper subgraph $A' \subseteq A$ such that $\bar{\Omega} \subseteq V(A')$, $H \cap A \subseteq A'$, $E(A') \neq \emptyset$ and for all $u, v \in \bar{\Omega}$, if u and v are joined by a path of A with no internal vertex in $\bar{\Omega}$ then they are joined by such a path in A'.*

Subproof. Suppose that such A' exists, and let $\mathscr{S}'$ be obtained from $\mathscr{S}$ by replacing (A, Ω) by (A', Ω) (since $E(A') \neq \emptyset$, (A', Ω) is not already a member of $\mathscr{S}$ and so $\mathscr{S}'$ is still a set rather than a "multiset"). Let G_1' be the graph such that $\mathscr{S}'$ is a 3-segregation of G_1', and let $G' = G_1' \cup G_2$. We claim that (G', Ω_1) has a crooked transaction of order $\geq \pi$. To see this, let $\mathscr{P}$ be a crooked transaction in (G, Ω_1) of order $\geq \pi$; then we may assume that $\mathscr{P}$ is not a crooked transaction in (G', Ω_1), and so there exists $P \in \mathscr{P}$ with $P \cap A \not\subseteq A'$. Since $V(\mathscr{P}) \subseteq \bar{\Omega}_1 \subseteq V(\mathscr{S})$, at least two vertices of P are in $\bar{\Omega}$; and hence P is unique, because $|\bar{\Omega}| \leq 3$. There is a unique subpath Q of P with ends in $\bar{\Omega}$, with $Q \subseteq A$ and with $Q \not\subseteq A'$; and it is easy to check that, from the assumptions about A' in (1), there is a path Q' of A' with the same ends as Q and with $\bar{\Omega} \cap V(Q') \subseteq \bar{\Omega} \cap V(Q)$. If we replace the subpath Q of P by Q', we obtain a crooked transaction in (G', Ω_1) of order $\geq \pi$, as claimed. We give the details, replacing P as a whole. Let B be the subgraph of G such that (A, B) is a separation of G with $V(A \cap B) = \bar{\Omega}$. Then $V(\mathscr{P}) \subseteq V(B)$, since $V(A) \cap V(\mathscr{S}) = \bar{\Omega}$. We claim that for every path R of G with both ends in $V(B)$, there is a path R' of $A' \cup B$ with the same ends such that $V(R') \cap V(B) \subseteq V(R) \cap V(B)$. We prove this by induction on $|V(R)|$. If $R \subseteq B$ we take $R' = R$. If some internal vertex w of R is in $\bar{\Omega}$, we apply induction to the two subpaths of R between its ends and w, and take for R' a path between the ends of R in the union of the two paths obtained. Otherwise $R \not\subseteq B$ and no internal vertex of R is in $V(A \cap B)$, and so $R \subseteq A$ and both ends of R are in $\bar{\Omega}$. By the assumption about A' in (1), there is a path R' of A' with the same ends as R and no internal vertex in $\bar{\Omega}$, and then $V(R') \cap V(B) \subseteq V(R) \cap V(B)$, as required. This proves our claim. Now every member of $\mathscr{P}$ that is not a subgraph of B has at least two vertices in $\bar{\Omega}$, and so P is the only such member, since $|\bar{\Omega}| \leq 3$. Let P' be a path of $A' \cup B \subseteq G'$ obtained from P by our claim. Then P' is disjoint from the other members of $\mathscr{P}$, since they are subgraphs of B and $V(P') \cap V(B) \subseteq V(P)$. Hence replacing P by P' gives a crooked transaction

in (G', Ω_1) with the same ends.[F10] But then from the inductive hypothesis, since $H \subseteq G'_1$, the result holds for G' and hence for G, as required. This proves that we may assume (1).

[p. 42] (2) *For each $(A, \Omega) \in \mathscr{S}$ either $A \subseteq H$ or A has at most one circuit.*

Subproof. Suppose first that $H \cap A$ has a circuit C. As in the proof of step (1) of (6.5), it follows that $|\bar{\Omega}| = 3$ and there are three disjoint paths of $H \cap A$ from $V(C)$ to $|\bar{\Omega}|$. But then by (1) (setting $A' = H \cap A$) we deduce that $A \subseteq H$ as required. We may assume then that $H \cap A$ is a forest. Let B be a maximal forest of A with $H \cap A \subseteq B$. If $E(B) = \emptyset$ then every edge of A is a loop, and by (1), $|E(A)| \leq 1$ as required. We may assume then that $E(B) \neq \emptyset$. By (1) (setting $A' = B$) we may assume that there exist $u, v \in \bar{\Omega}$ joined by a path of A with no internal vertex in $\bar{\Omega}$ but not joined by any such path in B. Since B is a maximal forest, there is a path P of B between u and v, but it has an internal vertex $w \in \bar{\Omega}$. Then $\bar{\Omega} = \{u, v, w\}$. Let $B = B_1 \cup B_2$ where $V(B_1 \cap B_2) = \{w\}$ and $E(B_1 \cap B_2) = \emptyset$. Since there is a path of A between u and v in ~~B~~$A \setminus w$, there is an edge e of A with one end in $B_1 \setminus w$ and the other in $B_2 \setminus w$.[F11] By (1) (setting A' to be the graph obtained by adding e to B) we deduce that $A \setminus e = B$ and so A has only one circuit. This proves (2).

Since every graph with at most one circuit can be drawn in the plane with all its vertices incident with the infinite region, we deduce from (2) and the rigidity of H that for each $(A, \Omega) \in \mathscr{S}$, the drawing $H \cap A$ in the disc $\alpha(A, \Omega)$ can be extended to an embedding of A in this disc, in such a way

[F10] **Missing proof.** The paper says only that it is easy to check that the path Q' exists. The argument above is the one used in the formalization.

[F11] **Proof gap.** First, "in $B \setminus w$" is a misprint for "in $A \setminus w$". Second, there is a gap: $B \setminus w$ may have more than two components, and then an edge e between $B_1 \setminus w$ and $B_2 \setminus w$ need not join u and v in $(B + e) \setminus w$. For example, if w has three neighbours in B, with components T_u, T_v, T_3 of $B \setminus w$ and $B_2 = T_v \cup T_3$, the edge e may join T_u and T_3. The proof can be completed by choosing B with more care. Among the maximal forests of A that contain $H \cap A$ and P, take B such that the component T_u of $B \setminus w$ that contains u is as large as possible. Suppose that an edge g of A joins T_u to a component T of $B \setminus w$ that contains neither u nor v. Then T has no vertex in $\bar{\Omega}$. The forest B has exactly one edge f between T and w: at least one, since otherwise $B + g$ would be a larger forest, and at most one, since B has no circuit. If $f \in E(H)$, then the edges of $H \cap A$ inside T or between T and w form a forest with at least one edge, and it has a leaf in T. This vertex is a leaf of $H \cap A$ not in $\bar{\Omega}$, which the rigidity of H does not allow. So $f \notin E(H)$, and replacing f by g in B gives a forest of the same kind with a larger component of u, which is impossible. Hence every edge of A that leaves T_u, other than at w, ends in the component T_v of $B \setminus w$ that contains v. A path of $A \setminus w$ from u to v leaves T_u, so some edge e of A joins T_u to T_v, and $B + e$ joins u and v by a path that avoids w. So Claim (2) holds as printed.

that each $v \in \bar{\Omega}$ is represented by the point $\alpha(v)$. It follows that we may assume, to simplify notation, that G_1 is a drawing in Σ.

Let Σ' be the closure of the component of $\Sigma - U(C_1)$ which includes $U(C_2)$. Then Σ' is a closed cylinder; let $A = G_1 \cap \Sigma'$, and let B be the unique subgraph of G_1 such that (A, B) is a separation and $V(A \cap B) = V(C_1)$. Then (G, Ω_1) is the composition of $(A \cup G_2, \Omega_0)$ with (B, Ω_1, Ω_0), and the latter is "rural" in the sense of [3]. By [3, theorem (9.1)], we deduce that since (G, Ω_1) has a crooked transaction of order $\geq \pi$, so does $(A \cup G_2, \Omega_0)$. Now $(A \cup G_2, \Omega_0)$ is the composition of (G_2, Ω_2) and (A, Ω_0, Ω_2), and the latter is a rural neighbourhood of breadth $\geq \pi + 5$, because C_1, C_2 have regional distance $\geq \pi + 4$. Because[F12] there are $2\pi + 8$ disjoint paths of A from X to $V(C_2)$ with no vertex in $V(C_2)$ except their end vertices, it follows from (8.2) that $(A \cup G_2, \Omega_0)$ has a crooked transaction $\mathscr{P}$ of order $\geq \pi$ with $V(\mathscr{P}) \subseteq X$, such that no member of $\mathscr{P}$ has any internal vertex in $\bar{\Omega}_0$. Then $\mathscr{P}$ is a crooked transaction in (G, Ω_0); and since for each $P \in \mathscr{P}$,

$$P \cap H \subseteq (A \cup G_2) \cap H = A \cap H = H \cap \Sigma'$$

and $U(P) \cap U(C_1) = \{u, v\}$ where P has ends uv, it follows that $\mathscr{P}$ satisfies the theorem. □

[p. 43] 9. Constant Radius Discs

In this section we prove the existence, given a tangle in a drawing as usual, of a circuit of almost constant radius around any centre, and with other useful properties as well. We begin with the following.

(9.1) *Let H be a 2-cell drawing in Σ, with radial drawing K. Let $\mathscr{T}$ be a respectful tangle in H of order θ, let κ be an integer with $1 \leq \kappa \leq \theta$, let $z \in A(K)$, and let Z be the union of $ins(W)$, taken over all closed walks W of K of length $< 2\kappa$ with $z \subseteq ins(W)$. Then there is a closed disc $\Delta \subseteq \Sigma$ with $Z \subseteq \Delta - bd(\Delta)$.*

Proof. By [5, theorem (8.10)], Z is simply-connected, and by (7.1), $Z \neq \Sigma$. Hence by [2, theorems (4.2) and (11.9)], $Z \subseteq \Lambda$ for some open disc

[F12] **Proof gap.** A gap in the proof: (8.2) needs paths from X to the inner boundary $\bar{\Omega}_2$ of the neighbourhood (A, Ω_0, Ω_2), and here $\Omega_2 = \alpha^{-1}(F_2)$ comes from the end F_2 of the cylinder. The $2\pi + 8$ paths stop at $V(C_2)$, and in general they do not reach $\bar{\Omega}_2$, so (8.2) does not apply as written. The proof can be completed by cutting also at C_2, and applying (8.2) to the part of A between $U(C_1)$ and $U(C_2)$, with a natural order of $V(C_2)$ in place of Ω_2. This part is a rural neighbourhood of breadth at least $\pi + 5$, because C_1, C_2 have regional distance at least $\pi + 4$, and the $2\pi + 8$ paths run from X to its inner boundary $V(C_2)$. The rest of the proof is unchanged, so (8.4) holds as printed.

$\Lambda \subseteq \Sigma$. Since Z is closed, there is a closed disc $\Delta \subseteq \Lambda$ with $Z \subseteq \Delta - bd(\Delta)$, as required. $\square$

(9.2) *Let H be a 2-cell drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H of order θ, with metric d. Let $z \in A(H)$ and let κ be an integer with $2 \leq \kappa \leq \theta - 3$. Then there is a $(\kappa+2)$-zone Λ around z such that $x \subseteq \Lambda$ for every $x \in A(H)$ with $d(z,x) < \kappa$.*

Proof. Let K be a radial graph of H. By (9.1) there is a closed disc $\Delta \subseteq \Sigma$ such that $ins(W) \subseteq \Delta - bd(\Delta)$ for every closed walk W of K with length $< 2\theta$ and with $z \cap ins(W) \neq \emptyset$. Let z' be the atom of K corresponding to z. Since $\theta \geq 3$, there is a closed walk W of K of length $< 2\theta$ with $z' \subseteq ins(W)$, and so $z' \subseteq \Delta - bd(\Delta)$.

(1) *For every $x \in A(H)$, if $d(z,x) < \theta$ then $x \cap (\Delta - bd(\Delta)) \neq \emptyset$, and $x \subseteq \Delta - bd(\Delta)$ unless x is a region of H and $d(z,x) \geq \theta - 2$.*

Subproof. Let x' be the atom of K corresponding to x. Then $x' \subseteq \Delta - bd(\Delta)$, and so $x \cap (\Delta - bd(\Delta)) \neq \emptyset$. If $x \not\subseteq \Delta - bd(\Delta)$ then $x \not\subseteq x'$, and so x is a region of H, and there is a region r' of K incident with x' with $r' \not\subseteq \Delta - bd(\Delta)$. Let $e \in E(H)$ with $e \subseteq r'$. Then $d(x,e) \leq 2$, and yet $d(z,e) = \theta$, since $r' \not\subseteq \Delta - bd(\Delta)$. Thus $d(z,x) \geq \theta - 2$, as required.

Let Z be the union of z' and $ins(W)$, taken over all closed walks W of K of length $< 2\kappa$ with $z' \subseteq ins(W)$. Then $Z \neq \emptyset$, since $z' \subseteq Z$, and $Z \subseteq \Delta - bd(\Delta)$, since $\kappa \leq \theta$.

(2) *$d(z,r) \leq \kappa$ and $r \subseteq \Delta - bd(\Delta)$ for every region r of H with $\bar{r} \cap Z \neq \emptyset$.*

[p. 44] *Subproof.* If there exists $v \in V(H) \cap Z$ incident with r, then $d(v,r) \leq 1$, and so

$$d(z,r) \leq d(z,v) + d(v,r) \leq (\kappa - 1) + 1 = \kappa.$$

If there is no such v, then since $\bar{r} \cap Z \neq \emptyset$, it follows that $\bar{r} \cap z' \neq \emptyset$, and hence $d(z,r) \leq 2 \leq \kappa$. Thus in either case $d(z,r) \leq \kappa \leq \theta - 3$, and so $r \subseteq \Delta - bd(\Delta)$ by (1). This proves (2).

From (2) and (5.2) we deduce that there is a circuit C of H with $U(C) \subseteq \Delta$, bounding an open disc $\Lambda \subseteq \Delta$ including Z, and such that every edge of C is incident with a region r of H satisfying $\bar{r} \cap Z \neq \emptyset$. Moreover, $x \subseteq \Lambda$ for every $x \in A(H)$ with $d(z,x) < \kappa$; for if x' is the corresponding atom of K, then $x' \subseteq Z \subseteq \Lambda$, and so $x \cap \Lambda \neq \emptyset$, and therefore $x \subseteq \Lambda$, as claimed. It remains to show that Λ is a $(\kappa+2)$-zone around z. To see this, let Y be the union of $ins(W)$ taken over all closed walks W of length $< 2(\kappa+3)$ with $z' \subseteq ins(W)$. We claim that $U(C) \subseteq Y$. Since Y is closed, it suffices to show that $e \subseteq Y$ for all $e \in E(C)$. Let $e \in E(C)$, and let r be the region

of H with $r \subseteq \Lambda$, incident with e. Then, by choice of C, $\bar{r} \cap Z \neq \emptyset$ and so $d(z,r) \leq \kappa$ by (2). Since $d(r,e) \leq 2$ it follows that $d(z,e) \leq \kappa + 2$. Hence Y includes the region of K including e, and hence includes e. This proves that $U(C) \subseteq Y$. Since Y is simply-connected by [5, theorem (8.10)], there is a closed disc $\Delta^* \subseteq Y$ with $bd(\Delta^*) = U(C)$. Since $\Delta^* \subseteq Y \subseteq \Delta$ and $\bar{\Lambda} \subseteq \Delta$, and the closed discs Δ^*, $\bar{\Lambda} \subseteq \Delta$ are both bounded by $U(C)$, it follows that $\Delta^* = \bar{\Lambda}$, and so $\bar{\Lambda} \subseteq Y$. Now let $x \in A(H)$ with $x \cap \bar{\Lambda} \neq \emptyset$. Then $x \cap Y \neq \emptyset$, and so $d(z,x) \leq \kappa + 2$. Hence Λ is a $(\kappa+2)$-zone as required. □

The next result is not needed for this paper, and is included just for future reference.

(9.3) *Let H be a 2-cell rigid drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H of order θ, with metric d. Let $z \in A(H)$, and let κ be an integer with $2 \leq \kappa \leq \theta - 6$. Then there is a $(\kappa+5)$-zone Λ around z such that $H \cap (\Sigma - \Lambda)$ is rigid and $x \subseteq \Lambda$ for every $x \in A(H)$ with $d(z,x) < \kappa$.*

Proof. By two applications of (9.2) (with κ and $\kappa+3$) there is a $(\kappa+2)$-zone Λ_1 around z such that $x \subseteq \Lambda_1$ for every $x \in A(H)$ with $d(z,x) < \kappa$, and there is a $(\kappa+5)$-zone Λ_2 around z such that $x \subseteq \Lambda_2$ for every $x \in A(H)$ with $d(z,x) < \kappa + 3$. Since Λ_1 is a $(\kappa+2)$-zone it follows that $\bar{\Lambda}_1 \subseteq \Lambda_2$. Choose a circuit C of H with $U(C) \subseteq \Lambda_2$, bounding an open disc $\Lambda \subseteq \Lambda_2$ with $\Lambda_1 \subseteq \Lambda$, and such that Λ is maximal. Then Λ is a $(\kappa+5)$-zone around z since $\Lambda \subseteq \Lambda_2$, and $x \subseteq \Lambda_1 \subseteq \Lambda$ for every $x \in A(H)$ with $d(z,x) < \kappa$, and it remains to show that $H \cap (\Sigma - \Lambda)$ is rigid. Suppose not; then by (6.3) there is a region r of H with $r \cap \Lambda = \emptyset$ such that $\bar{r} \cap \bar{\Lambda} \neq \emptyset$ [p. 45] is disconnected. But $r \subseteq \Lambda_2$, since $bd(\Lambda) \cap bd(\Lambda_2) = \emptyset$, and it is easy to see that this contradicts the maximality of Λ. We give the details. The proof of (6.3) uses its hypothesis only as follows: for a region r of H with $r \cap \Lambda = \emptyset$, distinct $v_1, v_2 \in V(C)$, and a line F_2 with ends v_1, v_2 and $F_2 - \{v_1, v_2\} \subseteq r$, it finds a line $L \subseteq \bar{r} \cap \bar{\Lambda}$ with ends v_1, v_2, that is, a path of C between v_1 and v_2 drawn in $\bar{r}$. Hence we may choose r so that there are such v_1, v_2, F_2, for which neither of the two paths P_1, P_2 of C between v_1 and v_2 satisfies $U(P_i) \subseteq \bar{r}$. Then $r \subseteq \Lambda_2$ as above, and so $F_2 \subseteq \Lambda_2$. For $i = 1, 2$, let $D_i \subseteq \bar{\Lambda}_2$ be the closed disc bounded by $F_2 \cup U(P_i)$. Now $F_2 \not\subseteq \bar{\Lambda}$, since $\bar{r} \cap \Lambda = \emptyset$ and $r \cap U(C) = \emptyset$. Suppose that $U(P_1) \not\subseteq D_2$ and $U(P_2) \not\subseteq D_1$. Since also $F_2 \not\subseteq \bar{\Lambda}$, the closed discs $\bar{\Lambda}$ and D_2 meet only in $U(P_2)$, and their union is a closed disc in $\bar{\Lambda}_2$ bounded by $F_2 \cup U(P_1)$. So it equals D_1, by uniqueness, which is impossible since it includes $U(P_2)$. Hence we may assume that $U(P_1) \subseteq D_2$. Then $U(C) \subseteq D_2$, and so $\bar{\Lambda} \subseteq D_2$, since $\bar{\Lambda}$ is the unique closed disc in $\bar{\Lambda}_2$ bounded by $U(C)$. Moreover $D_2 \subseteq \Lambda_2$, since $\mathrm{bd}(D_2) = F_2 \cup U(P_2) \subseteq \Lambda_2$.

Since $U(P_1) \not\subseteq \bar{r}$, some edge e_0 of P_1 satisfies $e_0 \not\subseteq \bar{r}$, and then $e_0 \cap \bar{r} = \emptyset$, since an edge that meets $\bar{r}$ is incident with r. Let $Z = \Lambda \cup e_0$. Then Z is non-empty and arc-wise connected, since $\Lambda \subseteq Z \subseteq \bar{\Lambda}$ and $\bar{\Lambda}$ is a closed disc. Also $Z \cap \mathrm{bd}(D_2) = \emptyset$, since Λ is disjoint from r and from $U(C)$, and e_0 is disjoint from r, from $V(H)$, and from the edges of P_2. Hence $Z \subseteq D_2 - \mathrm{bd}(D_2)$. Let s be a region of H with $\bar{s} \cap Z \neq \emptyset$. Then $s \neq r$, since $\bar{r} \cap Z = \emptyset$, and so s is disjoint from $\mathrm{bd}(D_2) \subseteq r \cup U(H)$. Since s is connected and meets the open set $D_2 - \mathrm{bd}(D_2)$, it follows that $s \subseteq D_2 - \mathrm{bd}(D_2)$. By (5.2), applied with D_2 in place of Δ, there is a circuit C' of H with $U(C') \subseteq D_2$, bounding an open disc $\Lambda' \subseteq D_2$ including Z. Then $U(C') \subseteq \Lambda_2$, $\Lambda' \subseteq \Lambda_2$, and $\Lambda_1 \subseteq \Lambda \subseteq \Lambda'$. But $e_0 \subseteq \Lambda' - \Lambda$, and so $\Lambda' \neq \Lambda$, contrary to the maximality of Λ.[F13]

Thus $H \cap (\Sigma - \Lambda)$ is rigid, as required. □

(9.4) *Let H be a 2-cell drawing in Σ, let $z \in A(H)$, and let $\mathscr{T}$ be a respectful tangle in H of order θ, with metric d. Let $1 \leq \kappa \leq \theta - 7$, and let Λ be a κ-zone around z, including every $x \in A(H)$ with $d(z,x) \leq 3$. Let H be a subgraph of G, let $\mathscr{S}$ be a $\mathscr{T}$-local 3-segregation of G, and let α be an arrangement of $\mathscr{S}$ in Σ compatible with H. Then there is a closed disc $\Delta \subseteq \Sigma$ such that $\Lambda \subseteq \Delta$, $bd(\Delta)$ is α-normal, $H \cap \Sigma - \Delta$ has a circuit, and $d(z,x) \leq \kappa + 3$ for all $x \in A(H)$ with $x \cap \Delta \neq \emptyset$.*

Proof. By (9.2) there is a $(\kappa+3)$-zone Λ_1 around z such that $x \subseteq \Lambda_1$ for every $x \in A(H)$ with $d(z,x) \leq \kappa$. Then $\bar{\Lambda} \subseteq \Lambda_1$, since Λ is a κ-zone. Let $C_1 = H \cap bd(\Lambda_1)$, $C_2 = H \cap bd(\Lambda)$. Now $d(z,x) \leq \kappa + 3 \leq \theta - 4$ for all $x \in A(H)$ with $x \subseteq \bar{\Lambda}_1$; and $x \subseteq \Lambda$ for all $x \in A(H)$ with $d(z,x) \leq 3$. By (7.7), there is an α-normal O-arc $F \subseteq \bar{\Lambda}_1$, bounding a closed disc $\Delta \subseteq \bar{\Lambda}_1$ with $\bar{\Lambda} \subseteq \Delta$. Then Δ satisfies the theorem. □

Let H be a 2-cell drawing in Σ, let $z \in A(H)$, and let $\mathscr{T}$ be a respectful tangle in H of order θ, with metric d. Let $\beta \geq 4$ be an even integer. A triple (Λ, C, X) is a *battlefield of size β around z* if

(i) Λ is a $(3\beta+3)$-zone around z, and $H \cap (\Sigma - \Lambda)$ is a rigid drawing in Σ

(ii) $X \subseteq V(H) \cap bd(\Lambda)$, and $\theta \geq 12\beta + 15$, and X is free with respect to the tangle of order $\theta - 4(3\beta+3) - 2$ obtained by clearing Λ,

(iii) C is a circuit of H with $U(C) \subseteq \Lambda$, and there are β disjoint paths of H from $U(C)$ to X, each with no vertex in $H \cap bd(\Lambda)$ except its last,

(iv) the regional distance between $U(C)$ and $bd(\Lambda)$ is at least $1 + \frac{1}{2}\beta$,

(v) the open disc in Λ bounded by $U(C)$ includes every $x \in A(KH)$

[F13] **Missing proof.** The paper says only that it is easy to see that r contradicts the maximality of Λ. The argument above is the one used in the formalization.

with $d(z,x) < \beta$.[F14]

The main result of this section is the following.

(9.5) *Let H be a 2-cell, rigid drawing in Σ, let $z \in A(H)$, and let $\mathscr{T}$ be a respectful tangle in H of order θ. Let $\beta \geq 4$ be an even integer with $\theta \geq 16\beta + 17$. Then there is a battlefield around z of size β.*

Proof. Let d be the metric of $\mathscr{T}$. By two applications of (9.2), with $\kappa = \beta$ and $\kappa = 3\beta + 1$, we deduce that there is a $(\beta+2)$-zone Λ_2 around z such that $x \subseteq \Lambda_2$ for every $x \in A(H)$ with $d(z,x) < \beta$, and a $(3\beta+3)$-zone Λ_1 around z such that $x \subseteq \Lambda_1$ for every $x \in A(H)$ with $d(z,x) \leq 3\beta$. Then $\bar{\Lambda}_2 \subseteq \Lambda_1$ because $\beta + 2 \leq 3\beta$, and $z \subseteq \Lambda_2$ because $d(z,z) = 0 < \beta$.

[p. 46] Now for $v_i \in V(H \cap bd(\Lambda_i))$ $(i = 1,2)$, $d(z,v_1) \geq 3\beta + 1$ since $v_1 \notin \Lambda_1$, and $d(z,v_2) \leq \beta + 2$ since $v_2 \in \bar{\Lambda}_2$, and so $d(v_1,v_2) \geq 2\beta - 1$. Since $d(u,v) \leq 2$ for any two vertices of H incident with the same region, it follows that the regional distance between $H \cap bd(\Lambda_1)$ and $H \cap bd(\Lambda_2)$ is at least β. From (5.5) there are $\beta - 1$ mutually disjoint circuits of H strictly between $H \cap bd(\Lambda_1)$ and $H \cap bd(\Lambda_2)$. Counting $H \cap bd(\Lambda_i)$ $(i = 1,2)$ as well, we deduce

(1) *There are $\beta+1$ mutually disjoint circuits $C_1, \ldots, C_{\beta+1}$ of $H \cap \bar{\Lambda}_1$, bounding closed discs $\Delta_1, \ldots, \Delta_{\beta+1} \subseteq \bar{\Lambda}_1$, respectively, such that $\bar{\Lambda}_2 \subseteq \Delta_1 \subseteq \Delta_2 \subseteq \cdots \subseteq \Delta_{\beta+1} \subseteq \bar{\Lambda}_1$.*

(2) *There are β mutually disjoint paths $P_1, \ldots, P_\beta$ of $H \cap \bar{\Lambda}_1$ from $bd(\Lambda_2)$ to $bd(\Lambda_1)$.*

Subproof. Suppose not. By a form of Menger's theorem for planar graphs, there is an O-arc $F \subseteq \bar{\Lambda}_1$ bounding a closed disc $\Delta \subseteq \bar{\Lambda}_1$ with $\bar{\Lambda}_2 \subseteq \Delta$, such that $F \cap U(H) \subseteq V(H)$ and $|F \cap V(H)| < \beta$; and we can choose F so that $F = U(C)$ for some circuit C of a radial graph of H. From (7.5), since Λ_1 is a $(3\beta+3)$-zone and

$$|E(C)| \leq 2(\beta - 1) < 2(\theta - (3\beta + 3)),$$

we deduce that $ins(C) \subseteq \bar{\Lambda}_1$; and hence $\bar{\Lambda}_2 \subseteq ins(C)$, and so $d(z,e) \leq \beta - 1$ for every edge e of $H \cap bd(\Lambda_2)$. But this contradicts the choice of Λ_2. This proves (2).

By [1, theorem (4.1)], or rather by the proof of that theorem, we deduce from (1) and (2) that (3) *There are circuits $C_1, \ldots, C_{\beta+1}$ as in (1), and paths $P_1, \ldots, P_\beta$ as in (2), such that $C_i \cap P_j$ is a path for $1 \leq i \leq \beta+1$ and $1 \leq j \leq \beta$.*

[F14] **Misprint.** A misprint: no radial drawing K appears in the definition, and d is the metric on $A(H)$. The proofs of (9.5) and (9.6) check and use this clause for $x \in A(H)$.

Let $h = \frac{1}{2}\beta + 2$; and let us choose $C_1, \ldots, C_{\beta+1}$ and $P_1, \ldots, P_\beta$ as in (3) such that Δ_h is maximal, where $\Delta_1, \ldots, \Delta_{\beta+1}$ are the closed discs in $\bar{\Lambda}_1$ bounded by $U(C_1), \ldots, U(C_{\beta+1})$ respectively.

(4) *$\bar{r} \cap \Delta_h$ is null or connected for every region r of H with $r \cap \Delta_h = \emptyset$.*

Subproof. Let $u, v \in \bar{r} \cap V(C_h)$; then there is a (unique) path $P(u,v)$ of C_h with ends u, v, and a line $F \subseteq r \cup \{u, v\}$ with ends u, v, such that the O-arc $F \cup U(P(u,v))$ bounds a disc Δ in Δ_{h+1} not including Δ_h. If possible, let us choose $u, v \in \bar{r} \cap V(C_h)$ in distinct components of $C_h \cap \bar{r}$, with $P(u,v)$ as short as possible; and let F, Δ be as above. Let $P = P(u,v)$. It follows that no internal vertex of P is incident with r, and no edge of P is incident with r (for if $e \in E(A)$ is incident with r then e has ends u, v, and so $P \subseteq C_h \cap \bar{r}$, contrary to the fact that u, v are in different components of $C_h \cap \bar{r}$).

[p. 47] Now by (6.2), $bd(\bar{r}) = U(C)$ for some circuit C of H; let Q be the path of C between u and v with $U(Q) \subseteq \Delta$. Then $P \cup Q$ is a circuit. Let $C'_1 = C_1, \ldots, C_{\beta+1} = C'_{\beta+1}$ except that C'_h is obtained from C_h by replacing P by Q. Let $P'_1 = P_1, \ldots, P'_\beta = P_\beta$ except that if P is a subpath of some P_j we replace P by Q. Since none of $U(C_1), \ldots, U(C_{\beta+1})$ intersects Δ except $U(C_h)$, it follows that $C'_1, \ldots, C'_{\beta+1}$ satisfy (1). Since none of $U(P_1), \ldots, U(P_\beta)$ meet $\Delta - U(P)$ (because each intersects C_h in a path) it follows that $P'_1, \ldots, P'_\beta$ satisfy (2). But also (3) is satisfied, as we can see as follows. We only need consider the intersections of $P'_1, \ldots, P'_\beta$ with C'_h. Thus, let $1 \leq j \leq \beta$. If P is a subpath of P_j then $P'_j \cap C'_h$ is obtained from $P_j \cap C_h$ by replacing P by Q, and so we may assume that it is not. If $P \cap P_j$ is null then $P'_j \cap C'_h = P_j \cap C_h$. We may assume then that $P \cap P_j$ is a path with one end v_1 or v_2 and the other internal to P. (Not both ends are internal to P, because only one edge of P_j in $\Delta_h - bd(\Delta_h)$ is incident with a vertex of C_h and $U(P_j) \cap (\Delta - U(P)) = \emptyset$.) But then $P'_j \cap C'_h$ is obtained from the path $P_j \cap C_h$ by deleting a terminal subpath, and so $P'_j \cap C'_h$ is a path. Thus (3) is satisfied. But this contradicts the maximality of Δ_h, and so proves (4).

We see that $\Lambda = \Delta_h - bd(\Delta_h)$ is a $(3\beta+3)$-zone around z, because $\Lambda \subseteq \Lambda_1$. Let $\mathscr{T}'$ be obtained by clearing Λ, where $\mathscr{T}'$ has order $\theta - 12\beta - 14$. By (4) and (6.3), $H \cap (\Sigma - \Lambda)$ is rigid. For $1 \leq j \leq \beta$, let v_j be the first vertex of P_j (starting from $bd(\Lambda_2)$) which is in $V(C_h)$, and let $X = \{v_1, \ldots, v_\beta\}$. Then there are β paths of H from $V(C_1)$ to X, mutually disjoint, and each meeting C_h just its last vertex (namely, the subpath of P_j from $V(C_1)$ to v_j, for $1 \leq j \leq \beta$). Moreover, Λ is a $(3\beta+3)$-zone around z, and Λ_1 is a $(\theta - \beta - 4(3\beta+3) - 2)$-zone around z, since $\theta - \beta - 4(3\beta+3) - 2 \geq 3\beta + 3$. Thus, by (7.8) (applied to $H \cap bd(\Lambda_1)$ and C_h, and with $\lambda = 3\beta+3$), it follows

that X is free with respect to $\mathscr{T}'$. Hence (Λ, C, X) satisfies the theorem. □

If α is an arrangement in a surface Σ of a segregation $\mathscr{S}$ of G, and $\Sigma' \subseteq \Sigma$ is a surface with $bd(\Sigma')$ α-normal, then for each $(A, \Omega) \in \mathscr{S}$, either $\alpha(A, \Omega) \subseteq \Sigma'$ or $\alpha(A, \Omega) \cap \Sigma' \subseteq bd(\Sigma') \cap bd(\alpha(A, \Omega))$. We define $\alpha^{-1}(\Sigma')$ to be the subgraph of G formed by

$$\bigcup (A : (A, \Omega) \in \mathscr{S} \text{ and } \alpha(A, \Omega) \subseteq \Sigma')$$

together with all $v \in V(\mathscr{S})$ such that $\alpha(v) \in \Sigma'$. We apply (9.5) to prove the following.

(9.6) *Let H be a 2-cell, rigid drawing in Σ, let $z \in A(H)$, and let $\mathscr{T}$ be a respectful tangle in H of order θ, with metric d. Let $\pi \geq 0$ with $\pi \neq 1, 3$, let $\beta \geq 2\pi + 8$ be even, and let $\theta \geq 16\beta + 17$. Let H be a subgraph of G, let J be the union of some bridges of H in G such that $d(z, v) < \beta$ for each* [p. 48] *$v \in V(H \cap J)$, and let G' be the union of H and all the other bridges of H in G. Let $\mathscr{S}$ be a $\mathscr{T}$-local 3-segregation of G', and let α be an arrangement of $\mathscr{S}$ in Σ compatible with H. Then there is a $(3\beta + 3)$-zone Λ around z, and a closed disc $\Delta \subseteq \Sigma$ with $bd(\Delta)$ α-normal and with $\Lambda \subseteq \Delta$, with the following properties:*

(i) *$d(z, x) \leq 3\beta + 6$ for all $x \in A(H)$ with $x \cap \Delta \neq \emptyset$*

(ii) *$x \subseteq \Delta - bd(\Delta)$ for all $x \in A(H)$ with $d(z, x) < \beta$*

(iii) *$H \cap \overline{\Sigma - \Delta}$ has a circuit, and $H \cap (\Sigma - \Lambda)$ is a rigid drawing in Σ*

(iv) *one of the following alternatives holds:*

(a) *there is a crooked transaction $\mathscr{P}$ of cardinality $\geq \pi$ in (G, Ω_0) (where Ω_0 is a natural order of $V(H) \cap bd(\Lambda)$ from $bd(\Lambda)$), such that $V(\mathscr{P})$ is free with respect to the tangle of order $\theta - 4(3\beta + 3) - 2$ obtained from $\mathscr{T}$ by clearing Λ, and each $P \in \mathscr{P}$ meets $H \cap (\Sigma - \Lambda)$ in precisely the ends of P*

(b) *there is no crooked transaction of cardinality $\geq \pi$ in $(\alpha^{-1}(\Delta) \cup J, \alpha^{-1}(bd(\Delta)))$.*

Proof. By (9.5), there is a battlefield (Λ, C, X) around z of size β. Hence Λ is a $(3\beta+3)$-zone. By (9.4) there is a closed disc $\Delta \subseteq \Sigma$ such that $\Lambda \subseteq \Delta$, $bd(\Delta)$ is α-normal, $H \cap \overline{\Sigma - \Delta}$ has a circuit, and $d(z, x) \leq 3\beta + 6$ for all $x \in A(H)$ with $x \cap \Delta \neq \emptyset$. Also, since (Λ, C, X) is a battlefield, it follows that $x \subseteq \Lambda \subseteq \Delta - bd(\Delta)$ for every $x \in A(H)$ with $d(z, x) < \beta$. Thus, Δ and Λ satisfy (i), (ii) and (iii), and it remains to verify (iv). Suppose that (iv)(b) is false; we shall show that (iv)(a) is true. Thus, $(G^*, \alpha^{-1}(bd(\Delta)))$ has a crooked transaction of cardinality $\geq \pi$, where $G^* = \alpha^{-1}(\Delta) \cup J$.

Now the regional distance between $U(C)$ and $bd(\Lambda)$ is at least $1 + \beta/2$. By (5.5), there is a circuit C_2 of H with $U(C_2) \subseteq \Lambda$, bounding an open disc

including $U(C)$, such that the regional distance between $U(C_2)$ and $bd(\Lambda)$ is at least $\beta/2$. By (7.7) applied to C and C_2, there is a closed disc Δ_2 with $bd(\Delta_2)$ α-normal, such that $U(C) \subseteq \Delta_2$ and Δ_2 is a subset of the closed disc in Λ bounded by $U(C_2)$. (Thus, we now have five "concentric" O-arcs; $bd(\Delta)$, $bd(\Lambda)$, $U(C_2)$, $bd(\Delta_2)$, $U(C)$.) Let $H \cap bd(\Lambda) = C_1$, and let Σ' be the closed cylinder $\Delta - (\Delta_2 - bd(\Delta_2))$. Let $H' = H \cap \Sigma'$. Since $H \cap \overline{\Sigma - \Delta}$ has a circuit and $H \cap \Delta_2$ has a circuit, it follows from (6.1) that H' is a rigid drawing in Σ'. Let $G_1 = \alpha^{-1}(\Sigma')$ and $G_2 = \alpha^{-1}(\Delta_2) \cup J$. Then (G_1, G_2) is a separation of G^*, and $G_1 \cap J$ is null, because for each $v \in V(H \cap J)$, $d(z, v) < \beta$ and hence $v \in \Delta_2 - bd(\Delta_2)$. Moreover, $(G^*, \alpha^{-1}(bd(\Delta)))$ is the composition of $(G_2, \alpha^{-1}(bd(\Delta_2)))$ and $(G_1, \alpha^{-1}(bd(\Delta)), \alpha^{-1}(bd(\Delta_2)))$. By (8.4) (applied to Σ', H', C_1, C_2) we deduce that (G^*, Ω_0) has a crooked transaction $\mathscr{P}$ of cardinality $\geq \pi$ such that for each $P \in \mathscr{P}$ with ends u, v say, $u, v \in X$ and $U(P \cap H') - \{u, v\}$ is included in the component of $\Sigma' - U(C)$ which includes $U(C_2)$. Since $U(P \cap H') - \{u, v\}$ is included in the component of $\Sigma' - U(C)$ which includes $U(C_2)$, and hence is included [p. 49] in Λ, and $U(G^* \cap H) \subseteq U(H') \cup \Lambda$, we deduce that $U(P \cap H) - \{u, v\} \subseteq \Lambda$. Consequently (iv)(a) holds, as required. □

10. The Main Proof

For an application in a future paper, let us extend (4.1) a little. (The following implies (4.1) by setting $\beta = 12$.)

(10.1) *Let H be a 2-cell, rigid drawing in Σ, and let $\mathscr{T}$ be a respectful tangle in H of order θ, with metric d. Let H be a subgraph of G, let $\beta \geq 12$ be an even integer with $\theta \geq 16\beta + 17$, and suppose that*

(i) *$d(u, v) < \beta$ for every H-path in G with ends u, v and*

(ii) *if $\mathscr{T}'$ is the tangle of order $\theta - 4(3\beta + 3) - 2$ obtained from $\mathscr{T}$ by clearing some $(3\beta + 3)$-zone Λ such that $H \cap (\Sigma - \Lambda)$ is a rigid drawing in Σ, then Λ is not an eye of $H \cap (\Sigma - \Lambda)$ in G with respect to $\mathscr{T}'$.*

Then there is a $\mathscr{T}$-local 3-segregation of G and an arrangement of it in Σ compatible with H.

Proof. We proceed by induction on $|V(G)| + |E(G)|$. If $G = H$ then the result is true, and so we may assume that there is a bridge J of H in G. Let G' be the union of H and all the other bridges of H in G. From the inductive hypothesis, there is a $\mathscr{T}$-local 3-segregation $\mathscr{S}$ of G' and an arrangement α of $\mathscr{S}$ in Σ compatible with H.

If $H \cap J$ is null, then we may replace some $(A, \Omega) \in \mathscr{S}$ by $(A \cup J, \Omega)$, thereby obtaining a 3-segregation of G satisfying the theorem. We assume then that there exists $u \in V(H \cap J)$. Let $z = \{u\}$. Now $d(z, v) \leq \beta - 1$

for every $v \in V(H \cap J)$, by hypothesis (i). By (9.6) with $\pi = 2$, there is a $(3\beta+3)$-zone Λ around z, and a closed disc $\Delta \subseteq \Sigma$ with $bd(\Delta)$ α-normal, satisfying (9.6)(i)–(iv).

Let $\mathscr{T}'$ be the tangle of order $\theta - (12\beta + 14)$ obtained from $\mathscr{T}$ by clearing Λ. From hypothesis (ii) applied to $\mathscr{T}'$, we deduce that (9.6)(iv)(a) is false, and so (9.6)(iv)(b) is true, that is, (G^*, Ω^*) has no cross, where $G^* = \alpha^{-1}(\Delta) \cup J$, $\Omega^* = \alpha^{-1}(bd(\Delta))$. Let $\mathscr{S}_1 = \{(A,\Omega) \in \mathscr{S} : \alpha(A,\Omega) \subseteq \Delta\}$, $\mathscr{S}_2 = \mathscr{S} - \mathscr{S}_1$. By (8.1) there is a 3-segregation $\mathscr{S}^*$ of G^* with $\bar{\Omega}^* \subseteq V(\mathscr{S}^*)$ and an arrangement α^* of $\mathscr{S}^*$ in Δ, such that $\alpha^*(v) = v$ for all $v \in \bar{\Omega}^*$. Since $H \cap \Delta$ is rigid by (6.1), it follows from (6.5) that we may choose α^* so that α^* is compatible with $H \cap \Delta$.

Let $(A,\Omega) \in \mathscr{S}^*$, and let (A,B) be the separation of G with $V(A \cap B) = \bar{\Omega}$. If $(B \cap H, A \cap H) \in \mathscr{T}$, then by (7.3), there exists $e \in E(A \cap H)$ with $d(z,e) \geq \theta - 3$, and yet $e \subseteq \Delta$, contrary to (9.6)(i). Thus $(B \cap H, A \cap H) \notin \mathscr{T}$, and since $\theta \geq 4$ and $(A \cap H, B \cap H)$ has order ≤ 3, it follows [p. 50] that $(A \cap H, B \cap H) \in \mathscr{T}$. Consequently, $\mathscr{S}' = \mathscr{S}_2 \cup \mathscr{S}^*$ is a $\mathscr{T}$-local 3-segregation of G. For $v \in V(\mathscr{S}')$, define $\alpha'(v) = \alpha(v)$ if $v \in V(\mathscr{S}_2)$, and $\alpha'(v) = \alpha^*(v)$ otherwise. For $(A,\Omega) \in \mathscr{S}'$, define $\alpha'(A,\Omega) = \alpha(A,\Omega)$ if $(A,\Omega) \in \mathscr{S}_2$, and $\alpha'(A,\Omega) = \alpha^*(A,\Omega)$ otherwise. Then α' is an arrangement of $\mathscr{S}'$ in Σ compatible with H, as required. □

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